

# ECE 105: Introduction to Electrical Engineering

Lecture 1

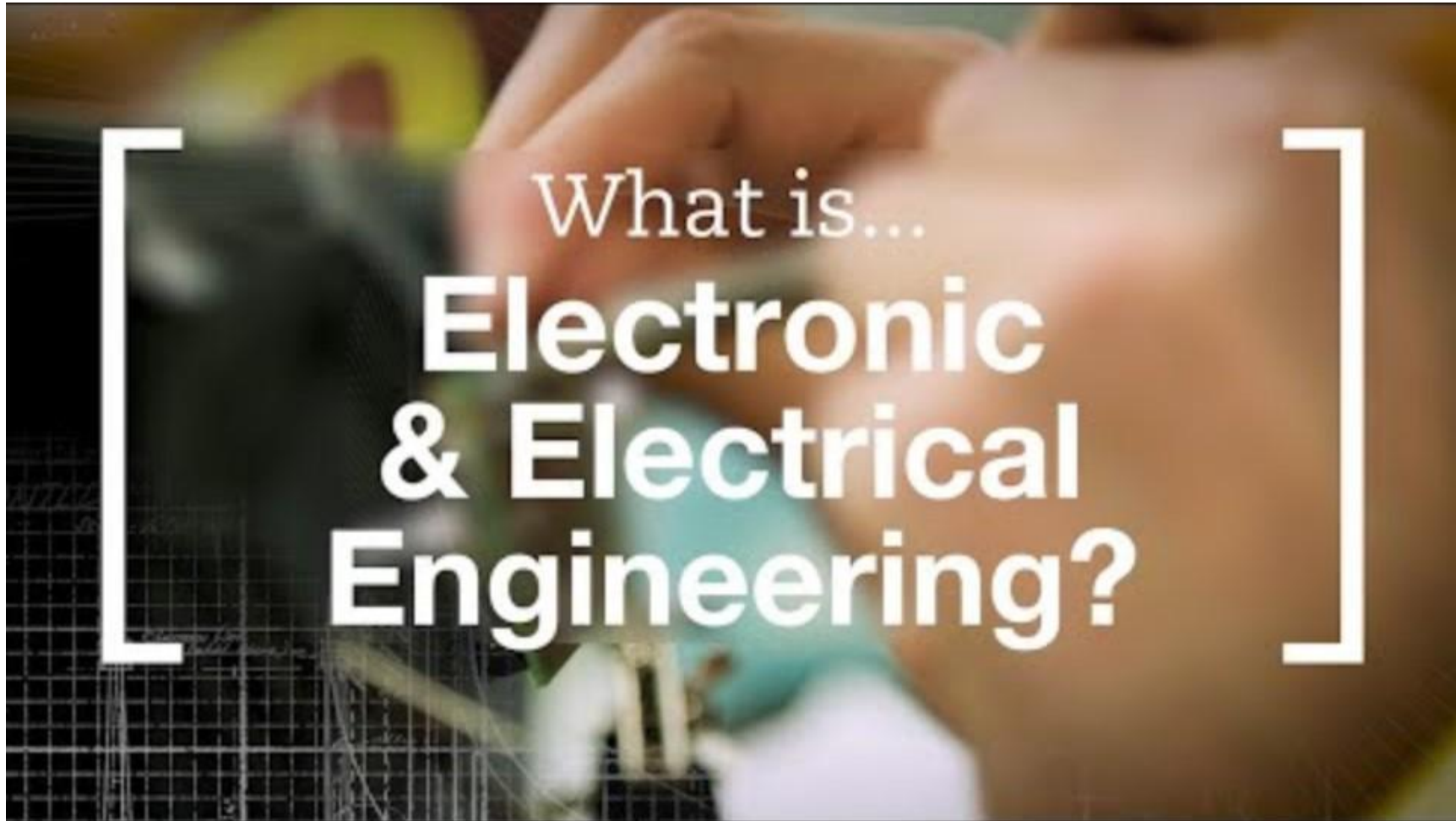
Introduction to EE, and ECE 105

Yasser Khan

Rehan Kapadia

# What is Electrical Engineering?

- Electrical Engineering is an engineering discipline concerned with the study, design, and application of equipment, devices, and systems which use electricity, electronics, and electromagnetism.



# Let's look at EE at play: 3 little pigs demo by Neuralink



# Neuralink – the neural implant

## LINK V0.9

1024 channels per Link

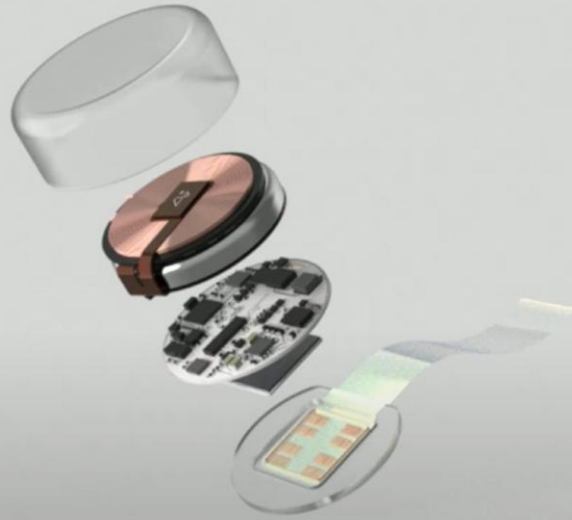
23 mm x 8 mm

Flush with skull (invisible)

6-axis IMU, temperature, pressure, etc.

Megabit wireless data rate, post compression

All day battery life

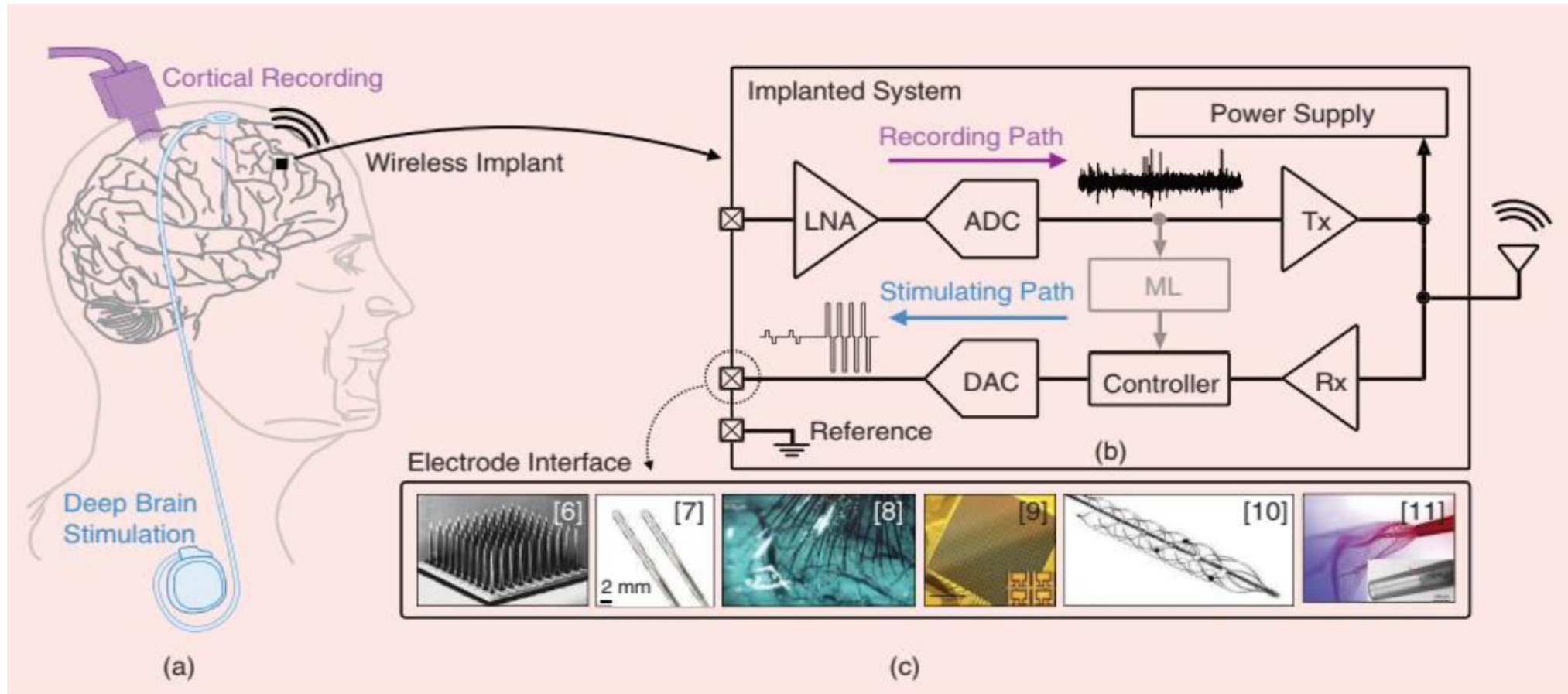


- Every step of the neural implant design and fabrication requires electrical engineering



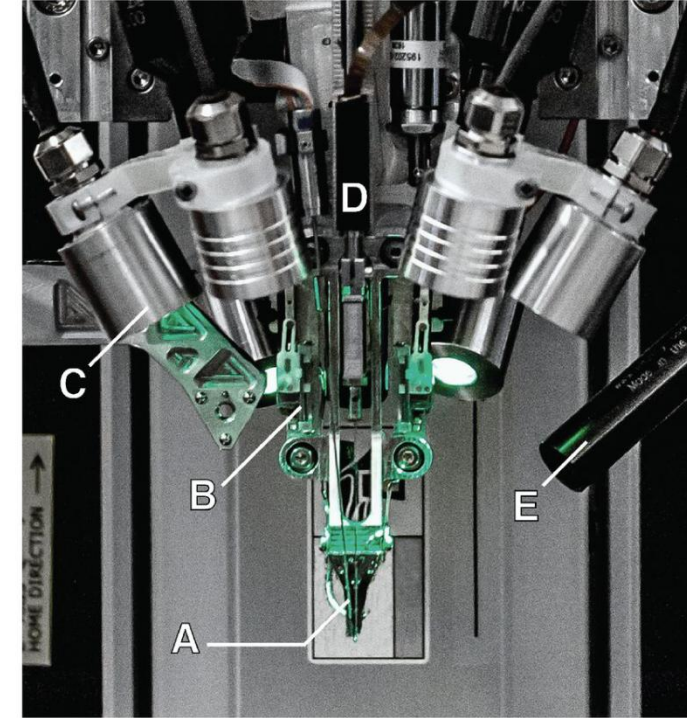
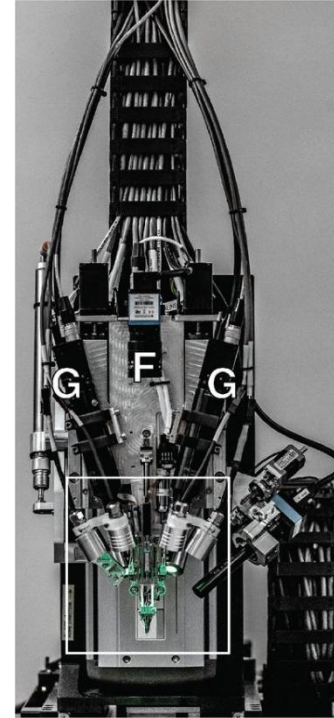
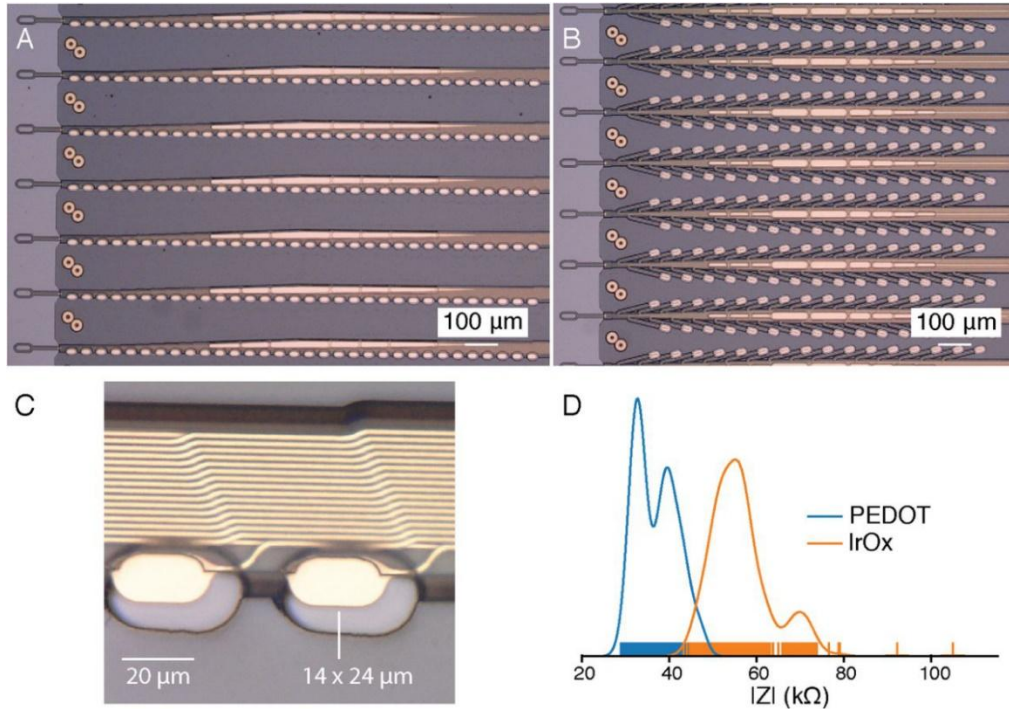


# Neural interfaces – under the hood



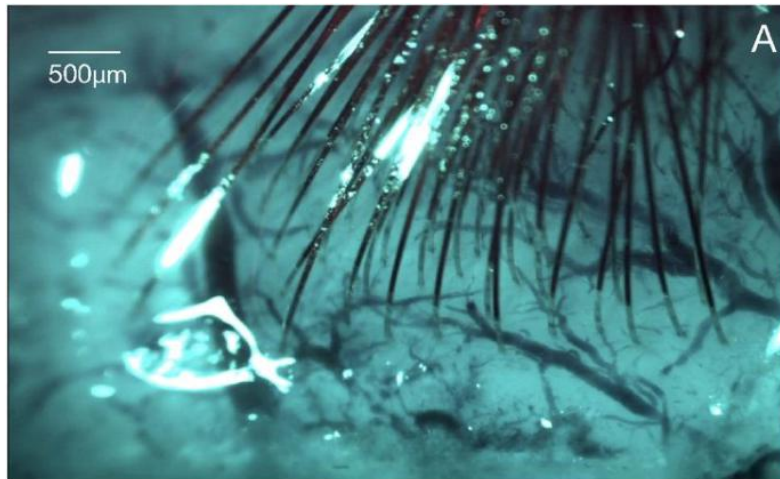
(a) Two clinical neural interfaces: a percutaneous neural recording implant, and a deep brain stimulator. (b) Components for an implantable bidirectional (black) or closed-loop (black and gray) wireless neural interface. (c) Example neural interface electrodes. [From left to right (adapted from Utah microelectrode array [6], SENSIGHT by Medtronic [7], ultrathin polymer threads by Neuralink [8], high-density neural matrix [9], Stentrode by Synchron [10], and ultraflexible syringe-injectable mesh [11]).] LNA: low-noise amplifier; ADC: analog-to-digital converter; Tx: transmitter; ML: machine learning; DAC: digital-to-analog converter; Rx: receiver.

Surgical robots need robotics, **electrical engineering**,  
mechanical engineering, control engineering



Sensors  
need

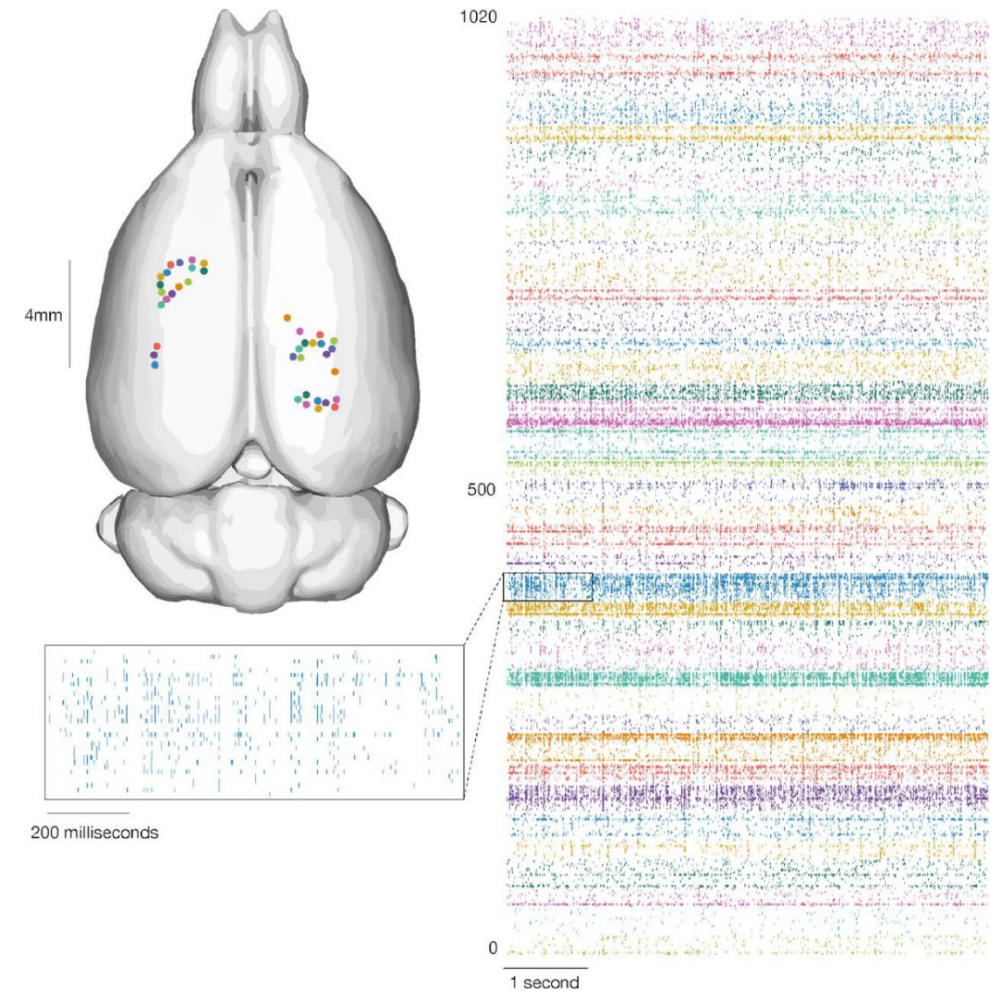
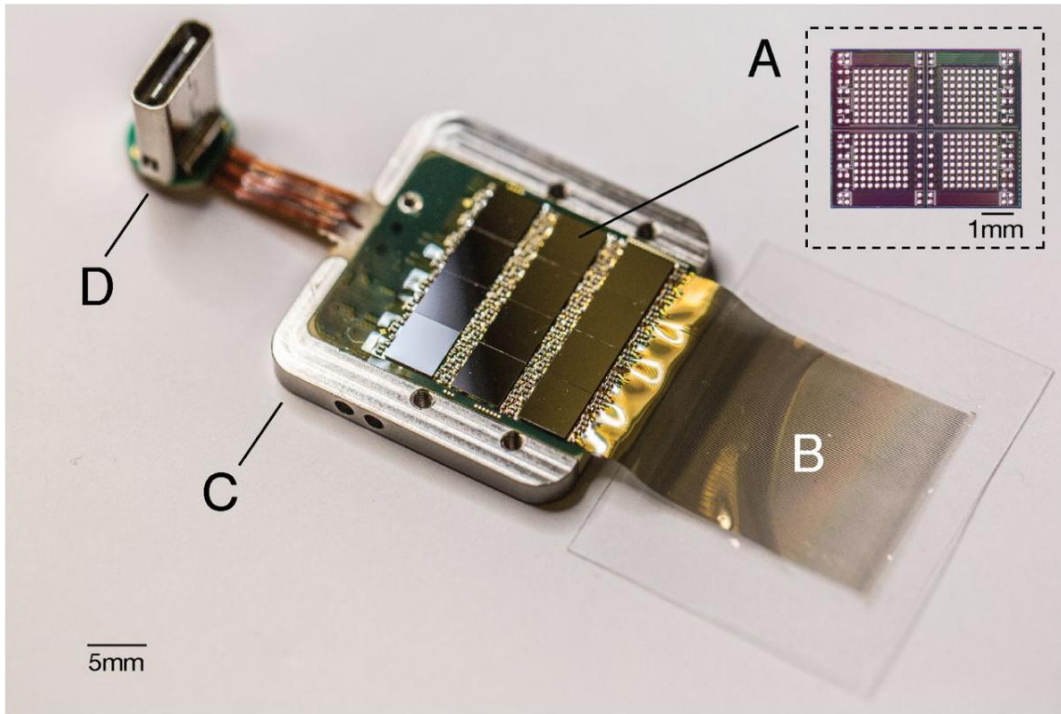
**electrical engineering**,  
mechanical engineering,  
materials science,  
chemical engineering



Implants need **electrical engineering**, computer science



# Brain recording from 1024 analog pixels

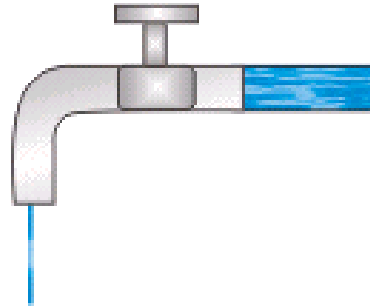
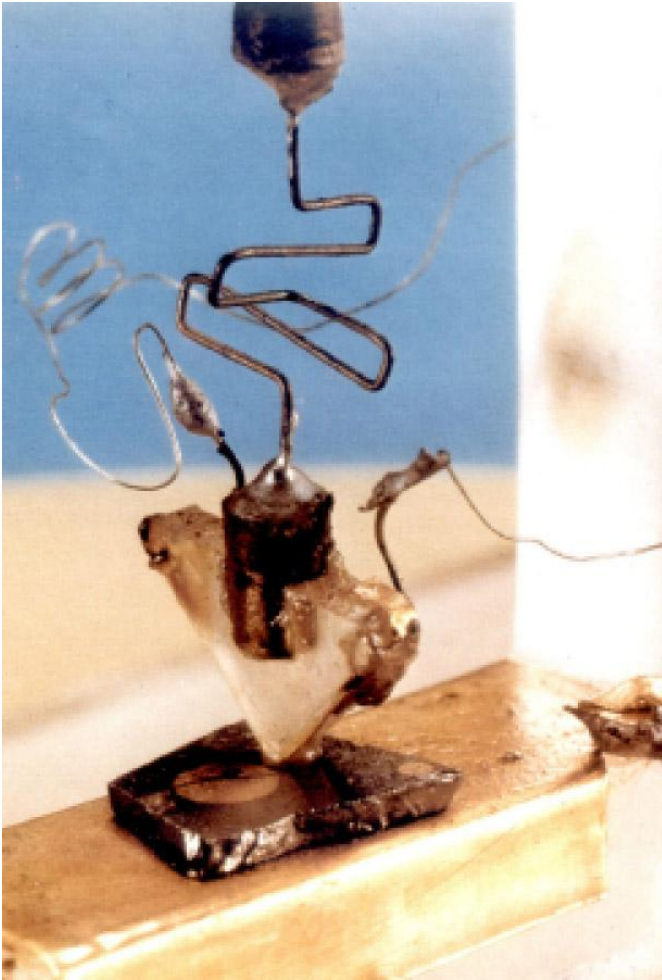


- Each neural recording unit requires sensors, integrated circuits, communications, and power

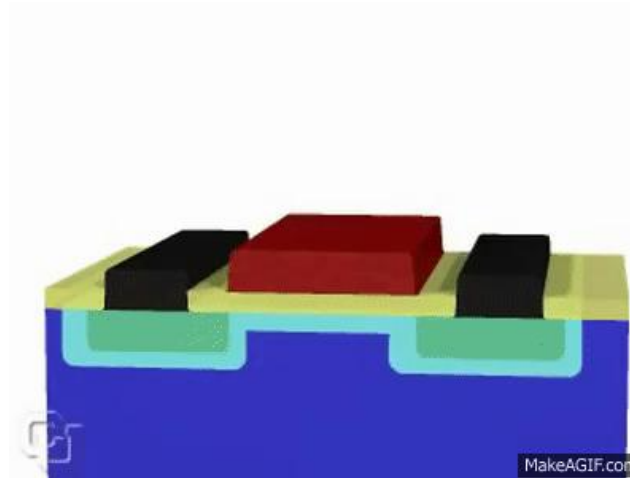
What is currently driving computing?



# How do we carry out so much computing?



We use  
transistors!



# 1956 Physics Nobel Prize

"for their researches on semiconductors and their discovery of the transistor effect"



**William Bradford Shockley**

🕒 1/3 of the prize

USA

Semiconductor  
Laboratory of Beckman  
Instruments, Inc.  
Mountain View, CA, USA



**John Bardeen**

🕒 1/3 of the prize

USA

University of Illinois  
Urbana, IL, USA



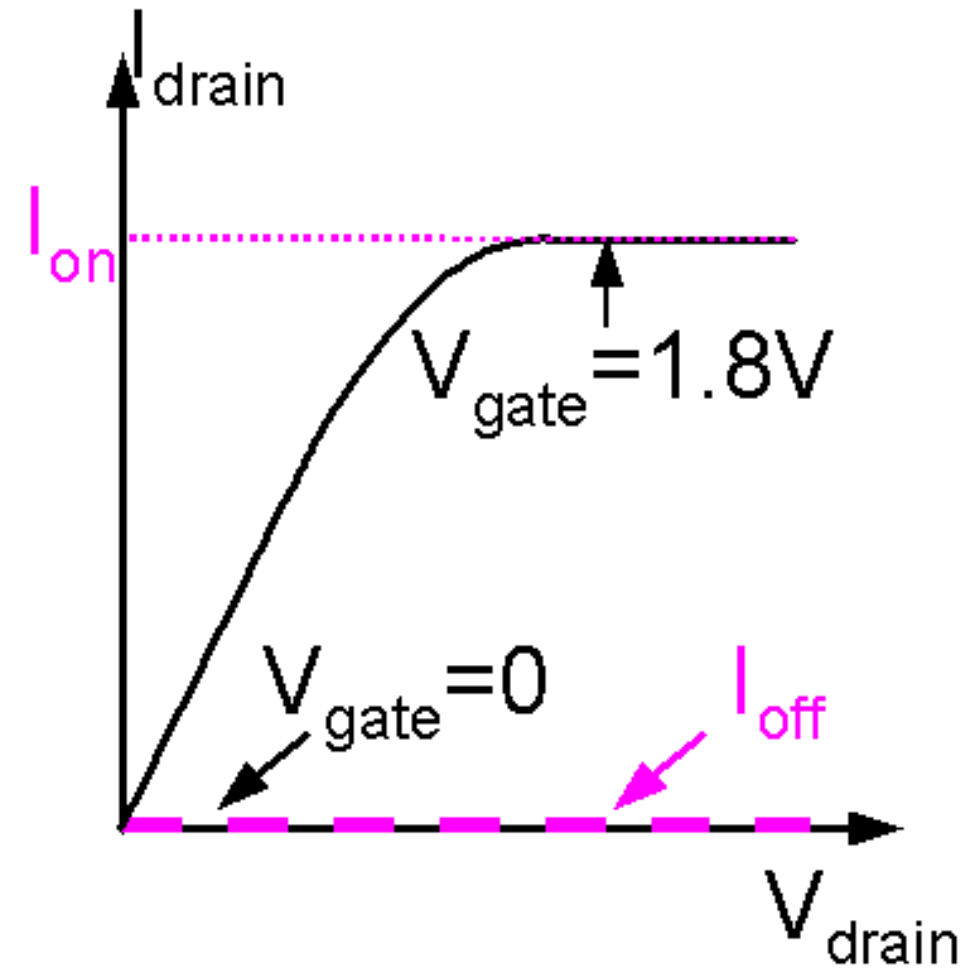
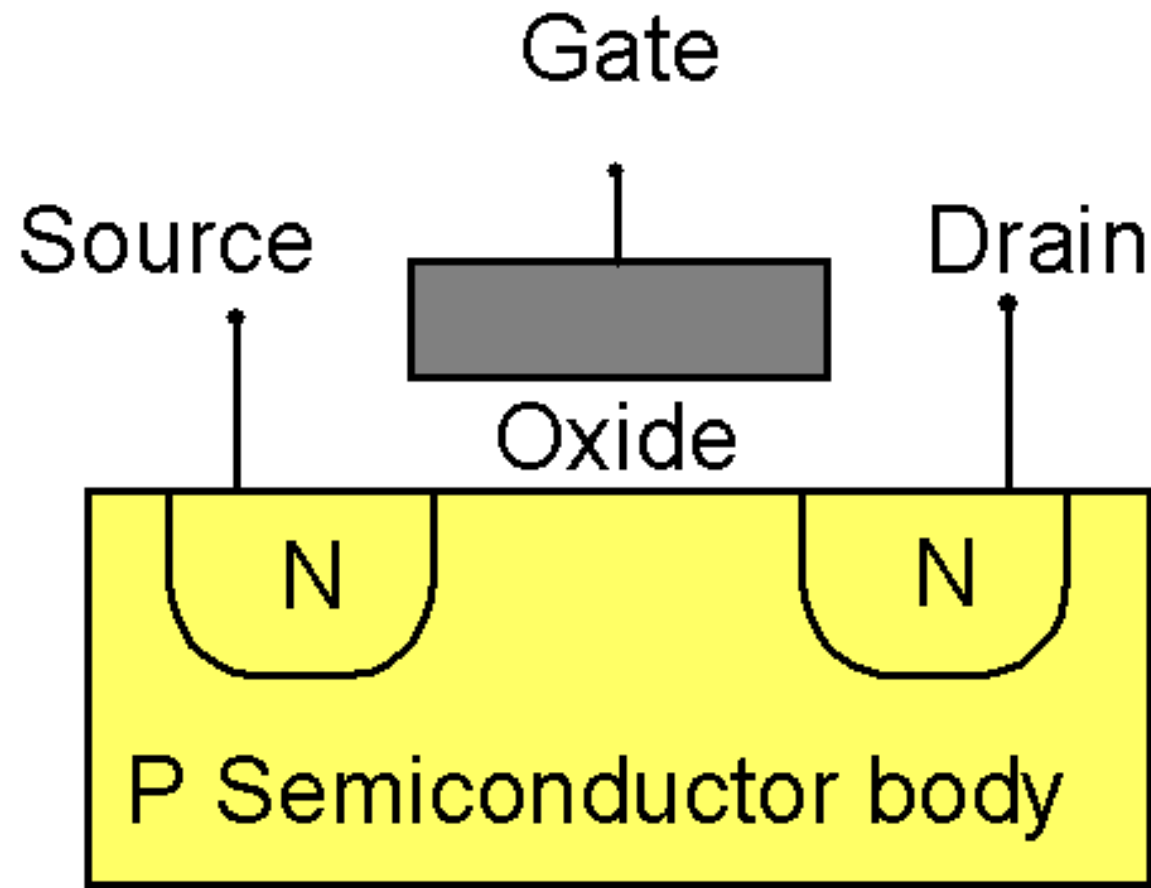
**Walter Houser Brattain**

🕒 1/3 of the prize

USA

Bell Telephone  
Laboratories  
Murray Hill, NJ, USA

# What is a transistor?



What is desirable: large  $I_{\text{on}}$ , small  $I_{\text{off}}$



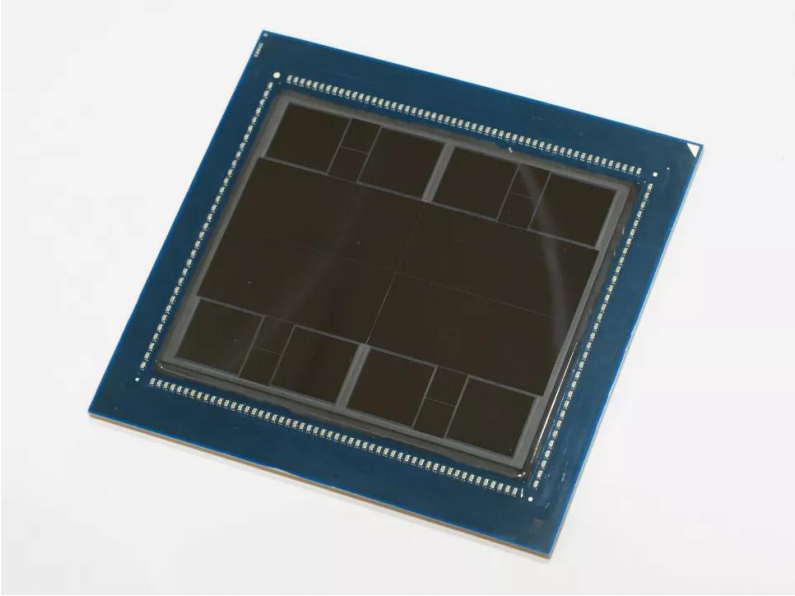
# What is a transistor?

1. It acts as an on-off switch.
2. We want to switch it on and off as fast as possible.
3. We want to switch it on and off with as little power usage as possible.

What is desirable: large  $I_{on}$ , small  $I_{off}$

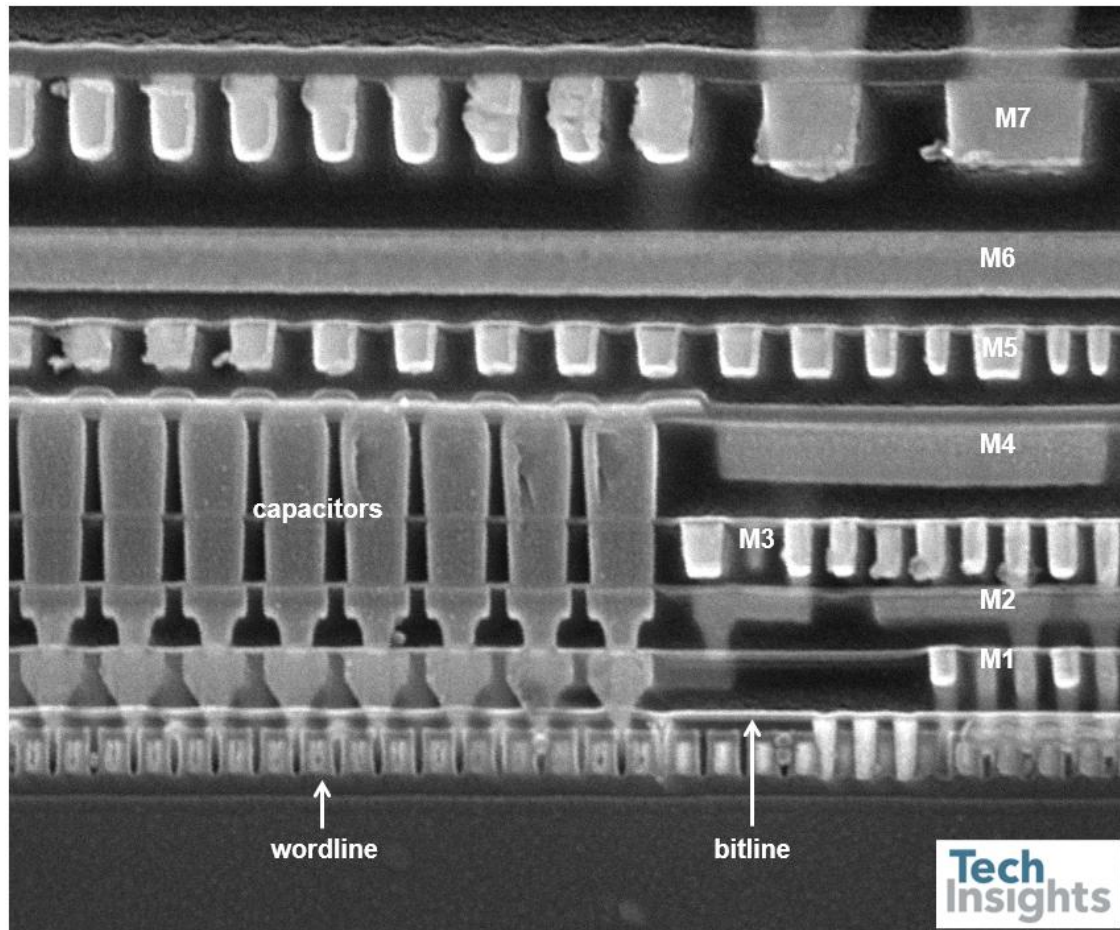
# What do modern day integrated circuits look like?

## AMD Instinct MI300 Accelerator



- Key points
  - Total of ~150 billion transistors
  - Uses 5 nm node technology
    - Density of transistors is ~130-230 million transistors/mm<sup>2</sup>
    - In the area of a cell , we could fit ~100k - 1 million transistors
- Multiple parts of this processor use 3-D stacked dies
  - Enables computing, memory, sensing, and stimulation to be integrated together
- Performance ~8x improved over previous MI250 Accelerator
  - If they upgraded the fastest supercomputer to this part, they could train the largest AI/ML systems

# What do these look like?



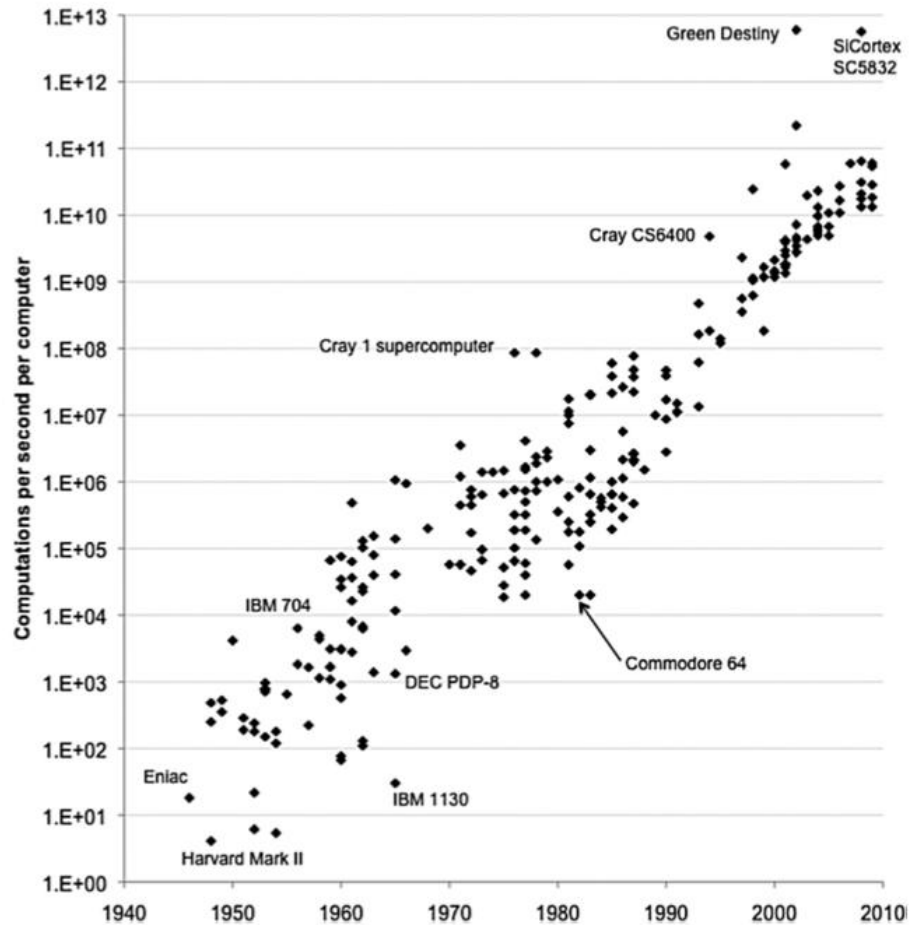
Wires to connect all the transistors together

Active devices (silicon transistors)

How did we get here?

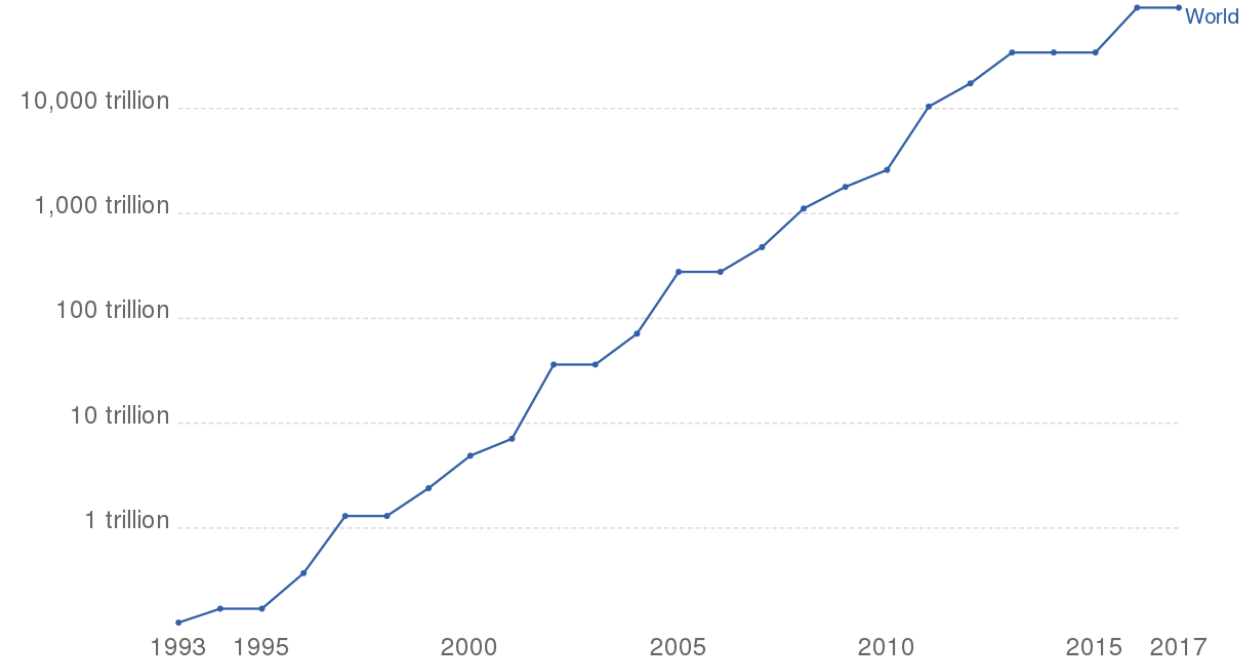


# How much has computing power increased?



## Supercomputer Power (FLOPS)

The growth of supercomputer power, measured as the number of floating-point operations carried out per second (FLOPS) by the largest supercomputer in any given year. (FLOPS) is a measure of calculations per second for floating-point operations. Floating-point operations are needed for very large or very small real numbers, or computations that require a large dynamic range. It is therefore a more accurate measure than simply instructions per second.



Source: TOP500 Supercomputer Database

- From 1990 to today, the worlds largest super computers have increased computing power by 100,000x
  - That's a lot!

# The First Moon Landing Was Achieved With Less Computing Power Than a Cell Phone or a Calculator



On board Apollo 11 was a computer called the Apollo Guidance Computer. It had 2,048 words of memory that could be used to store “temporary results”—data that is lost when there is no power. This type of memory is referred to as RAM (random access memory). Each word comprised 16 binary digits (bits), with a bit being a zero or a one. This means that the Apollo computer had **32,768 bits of RAM memory**.

My computer has 64,000,000,000 bits of RAM memory

That is 2,000,000 times more than what was needed in Apollo 11.

How do we make transistors faster?

# Scaling – How much have we scaled devices?

- 1971: The first commercial microprocessor...Intel 4004
  - 4-bits
  - Less than 1 square inch in area
  - Transistors that were 10 micrometers long
  - ~750 kilohertz speed
  - ...but more powerful than the computers from earlier that weighed 30 tons

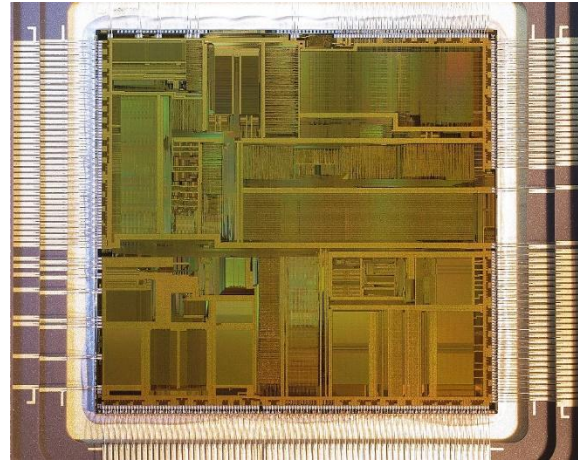


Let's say that this is a scaled version of how long a transistor was in 1971



# Scaling – How much have we scaled devices?

- 1990s: The first Pentium processor
- 66 MHz. That's 100 times faster than in 1970.
  - Can you think of anything else that has improved by 100x in 20 years?

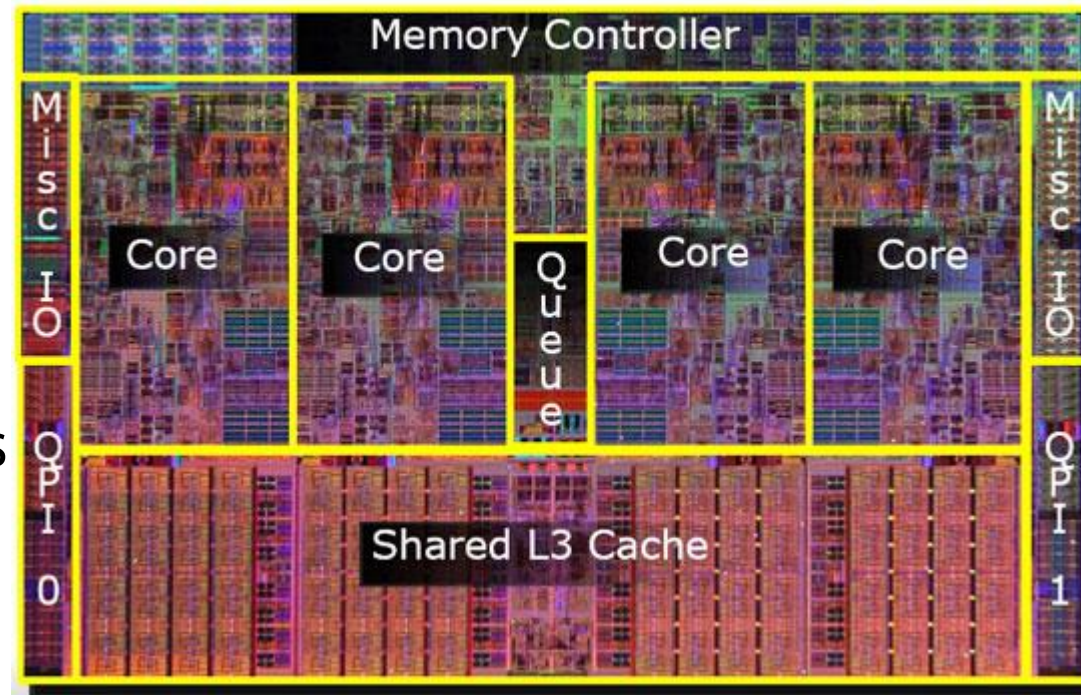


1971

1993

# Scaling – How much have we scaled devices?

- 2012:
  - 1-3 GHz
  - Multi-core
  - Super-fast
  - Each transistor is almost 1000 times smaller!

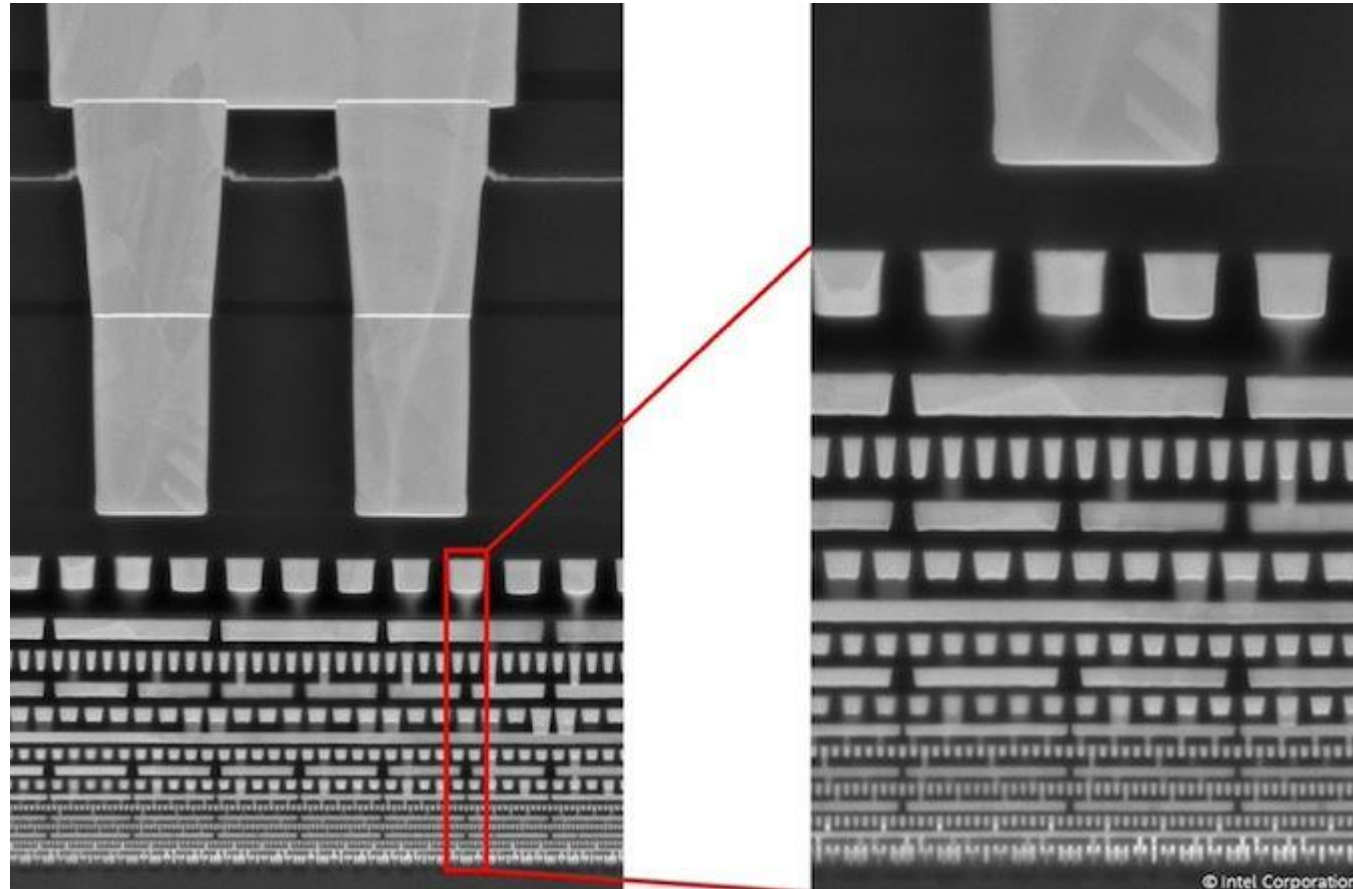


1971

1993

Can't even see the size on the screen...not enough pixels

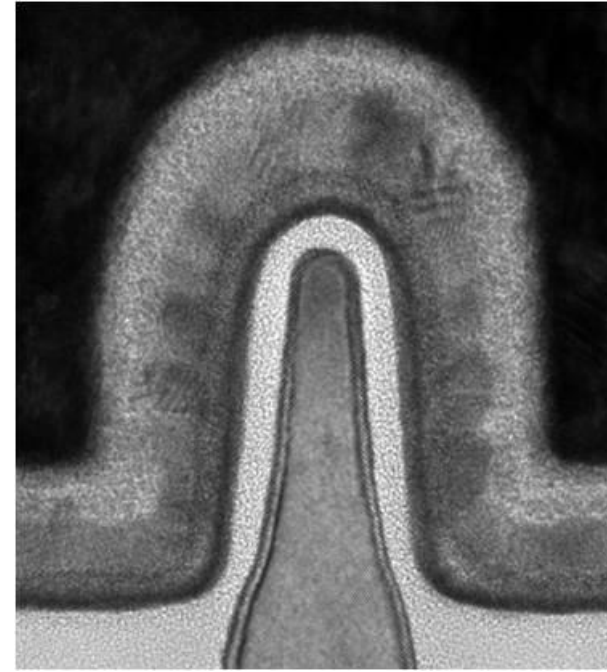
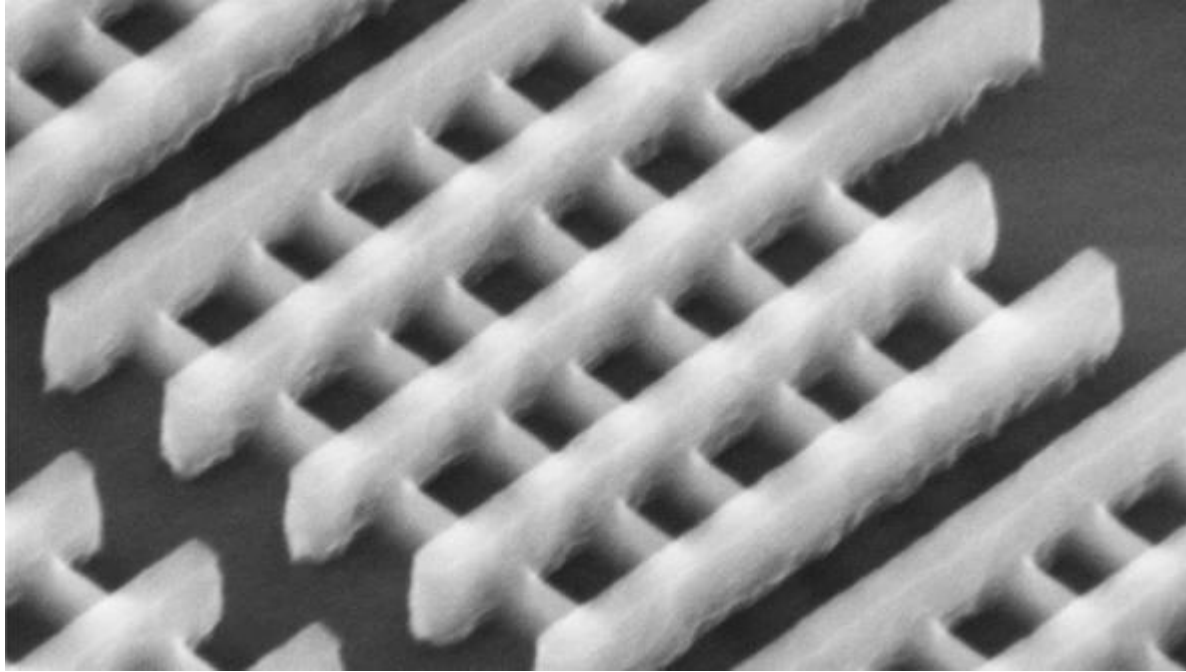
# Scaling – How much have we scaled devices?



Intel 4 cross section: to be released 2023

- What does a modern CPU cross section look like?
  - Where are the transistors?
  - What are we looking at?

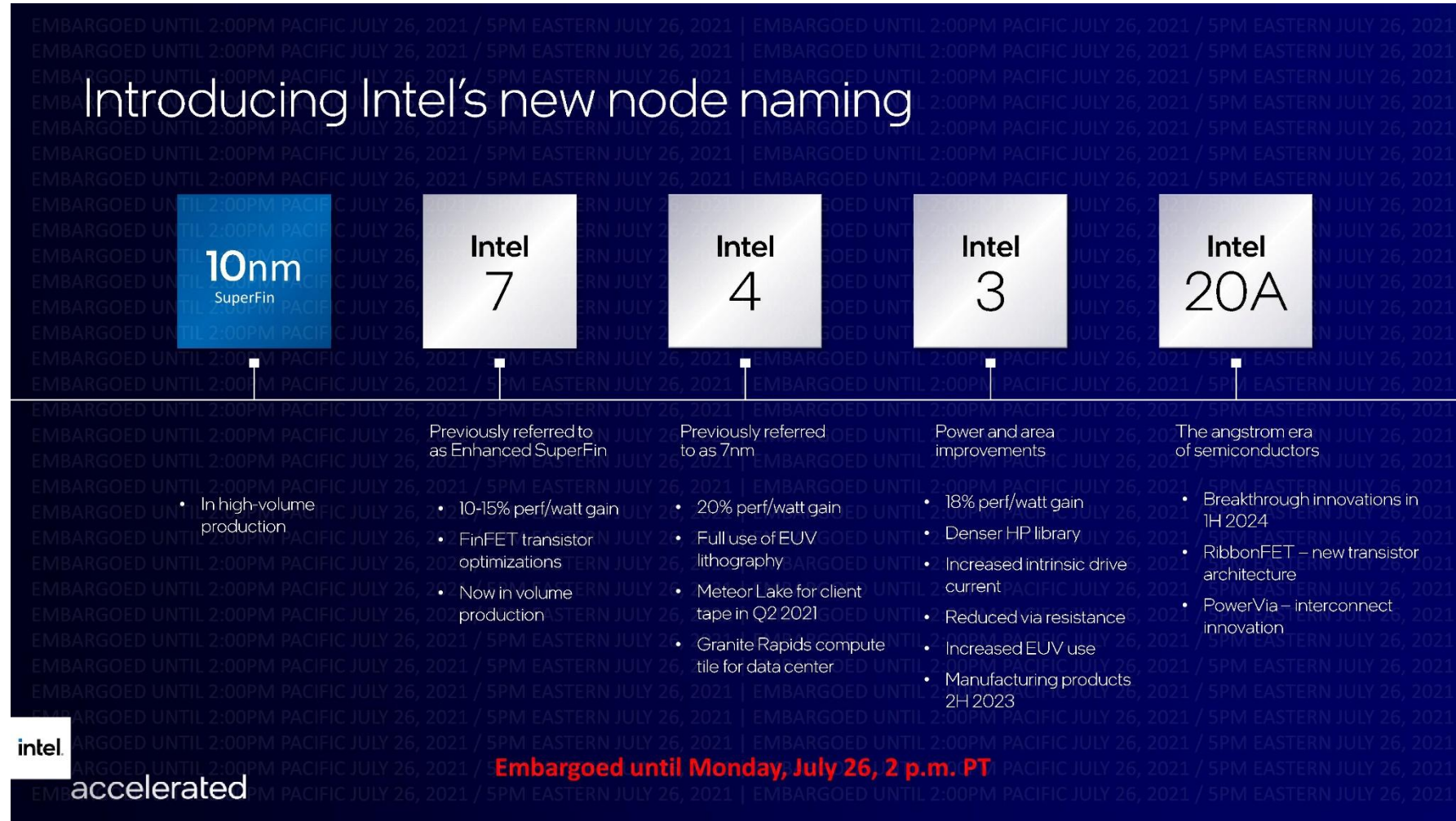
# FinFET Transistors (May 2011 and After)



- FinFETs
  - Why is it called that?
  - What do you think the advantage is?



# Scaling – How much have we scaled devices?

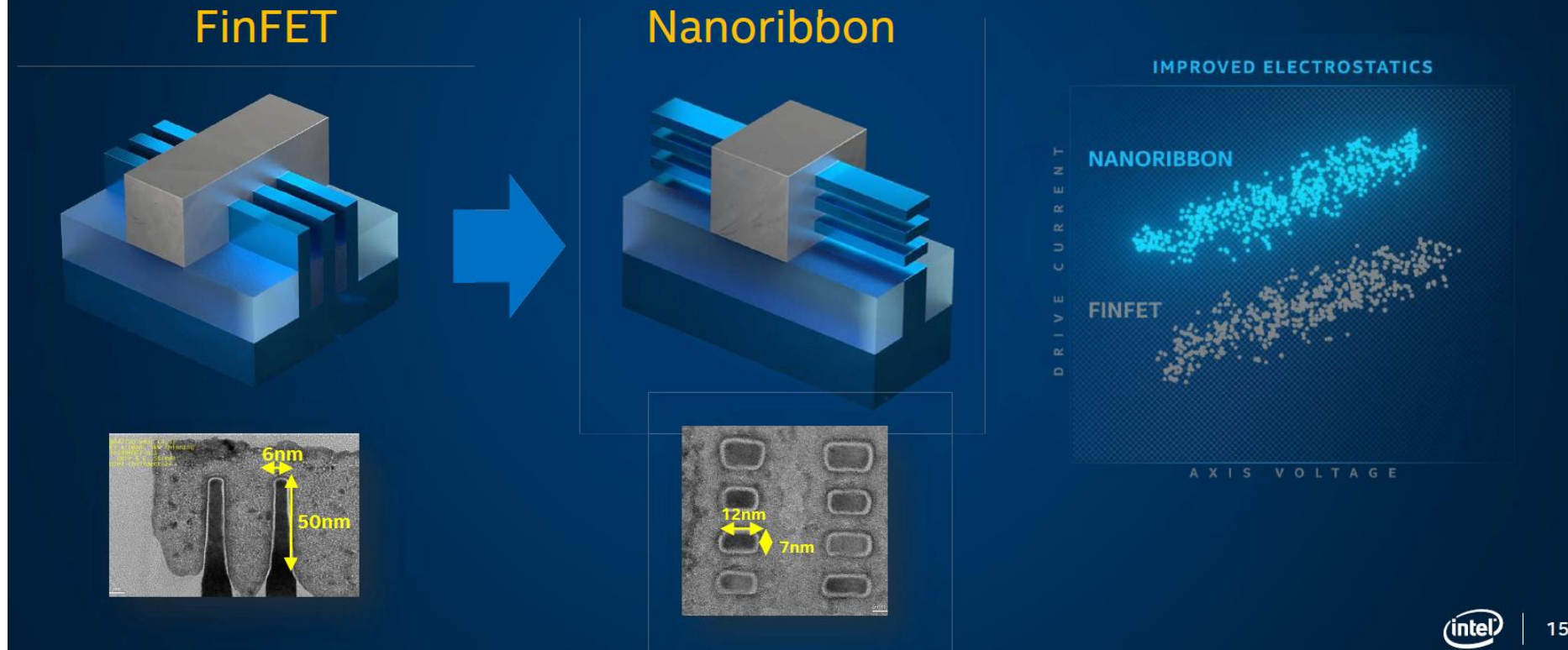


- This is a recent roadmap for Intel's transistor scaling plans



# Scaling – How much have we scaled devices?

## FINFET CHANGING TO NANOWIRE/NANORIBBON

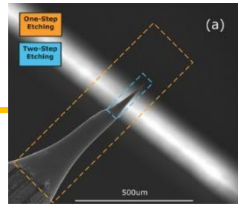


- What is a Nanoribbon FET?
  - Why does it give better performance?

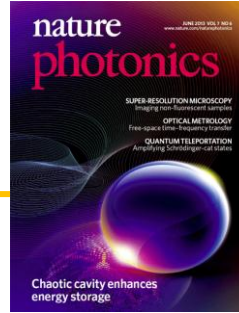
# My background and research



BS, EE, 2010



Electrical Engineering  
and Materials Science



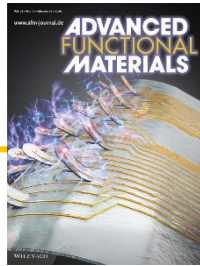
MS, EE, 2012

Started PhD, 2013

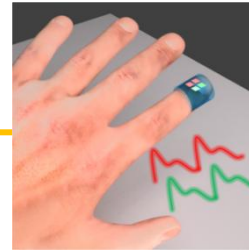


Bioelectronic and  
Biophotonic  
Systems

Flexible Hybrid  
Electronics  
2016



2015

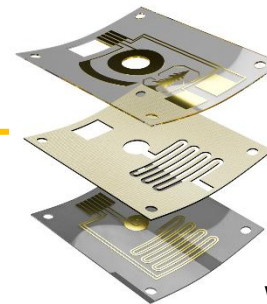


2014

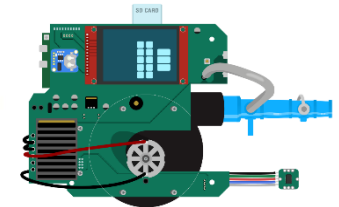
Stanford

Started  
postdoc, 2019

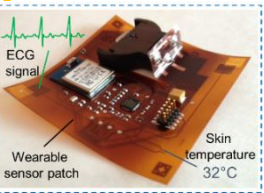
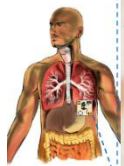
PhD,  
EECS,  
2018

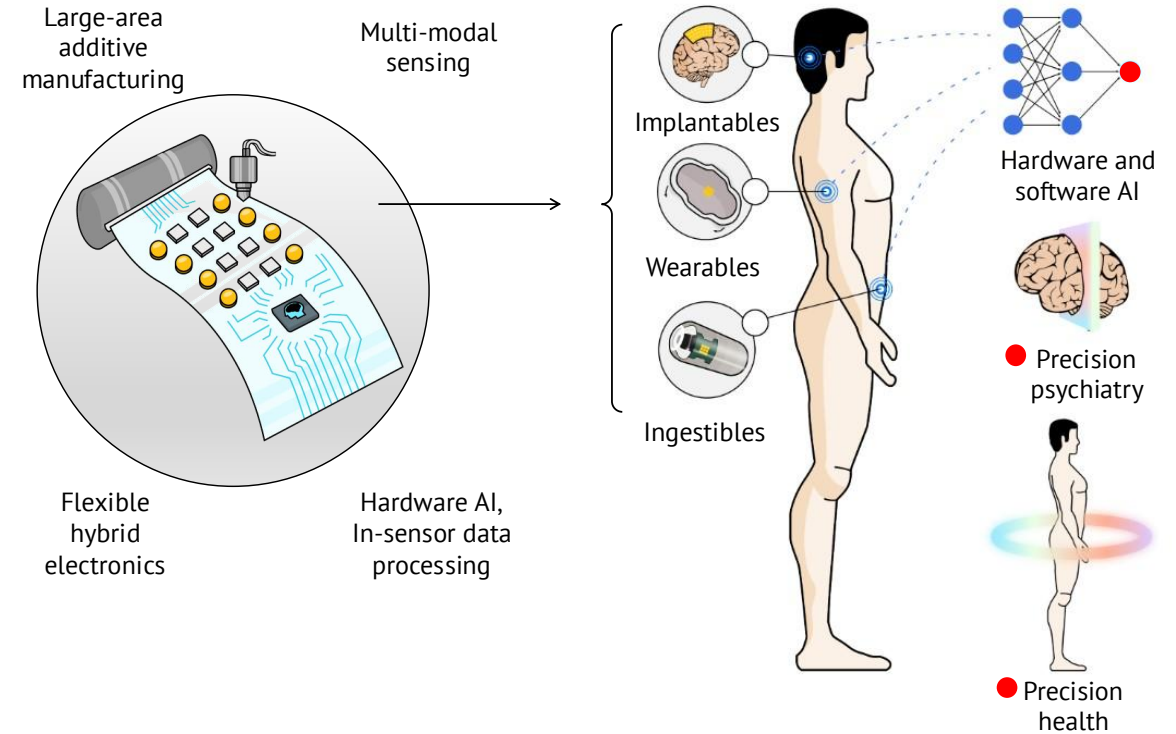


Wearables for  
precision health  
and psychiatry

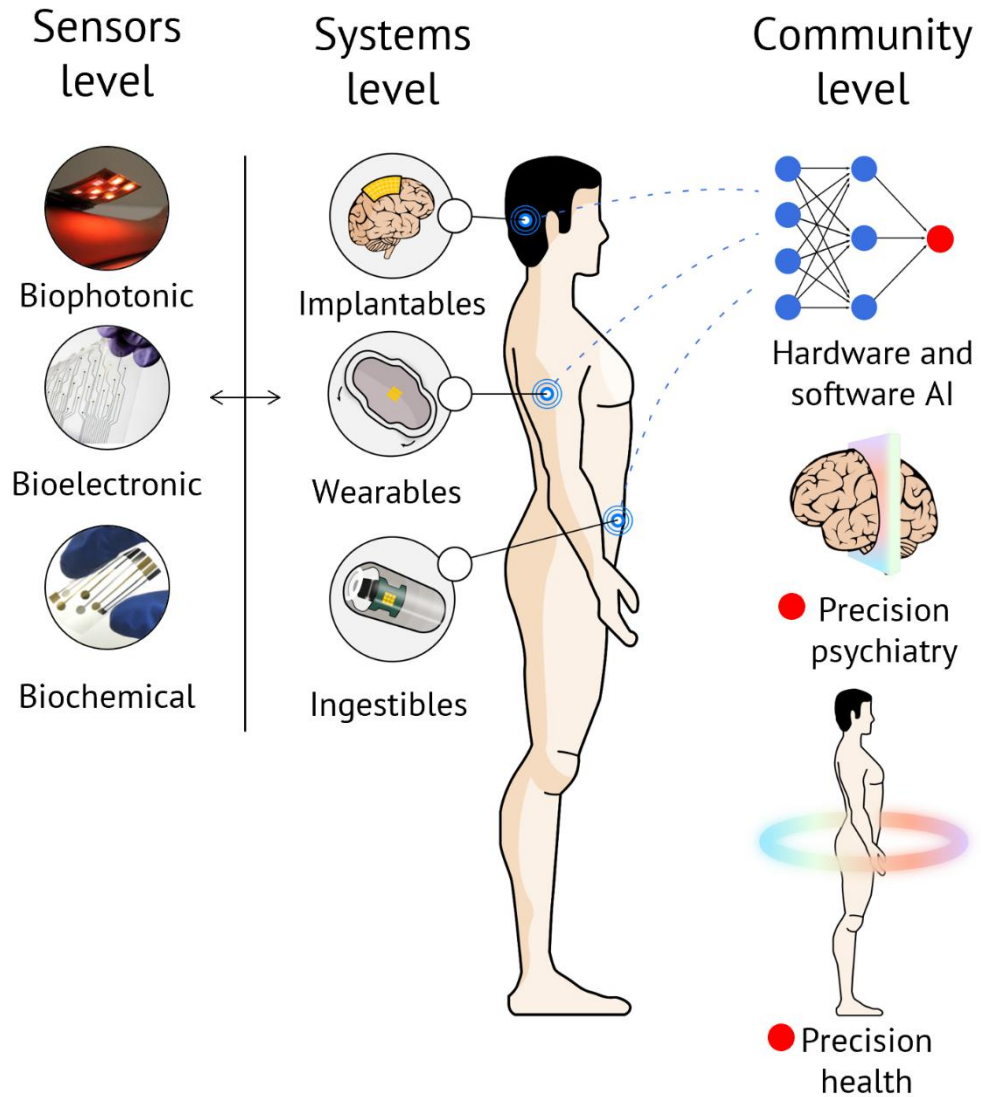


Ventilator for  
COVID-19

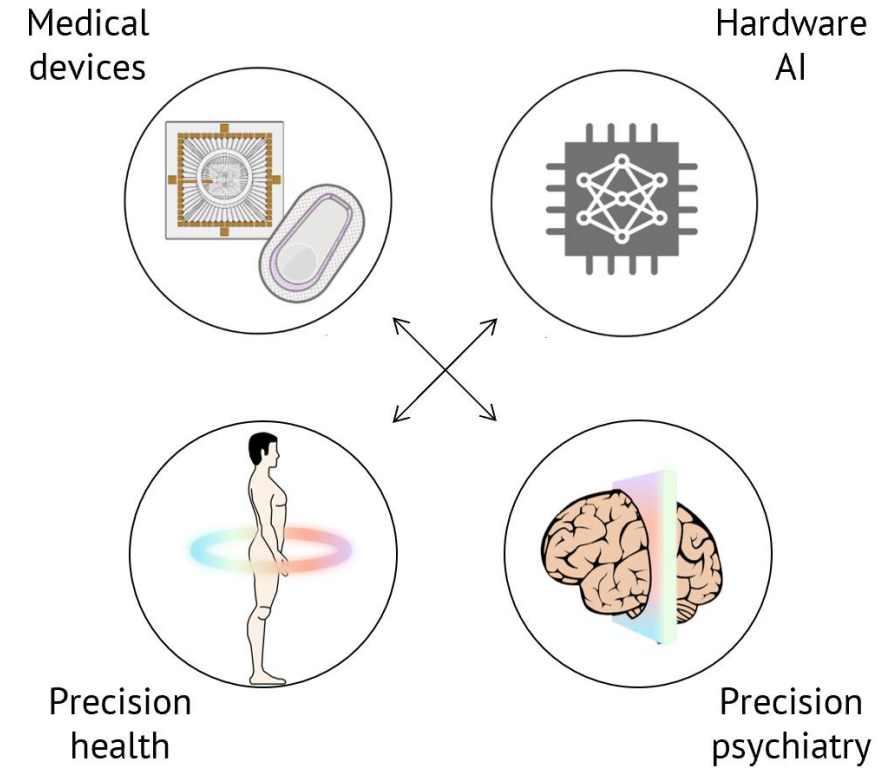






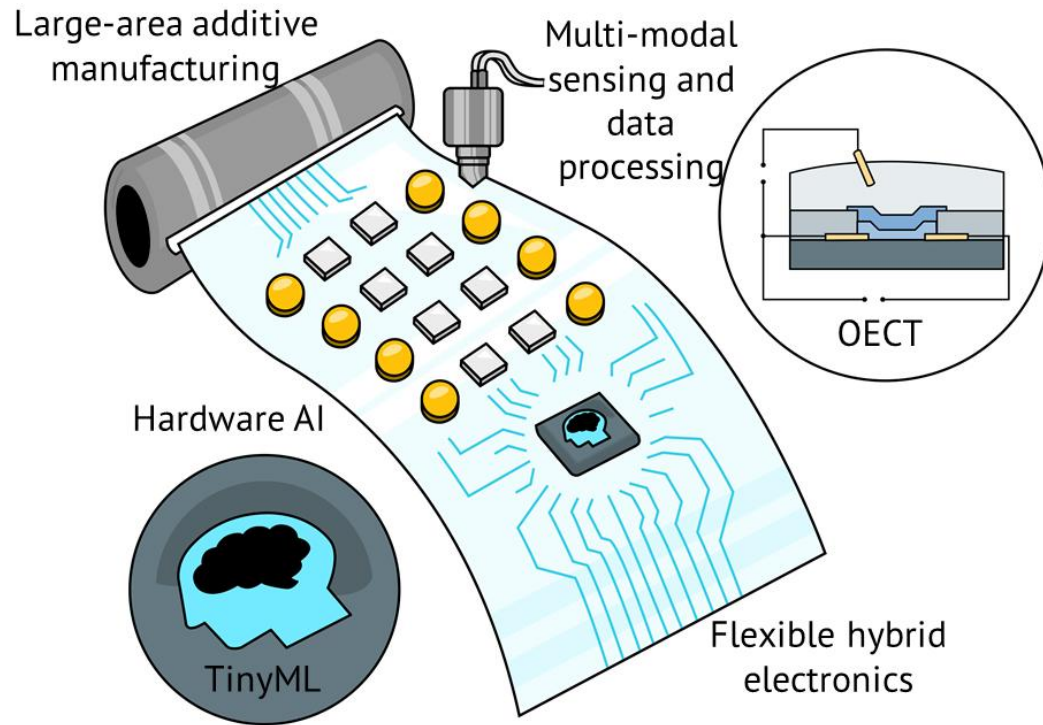


## Research Focus

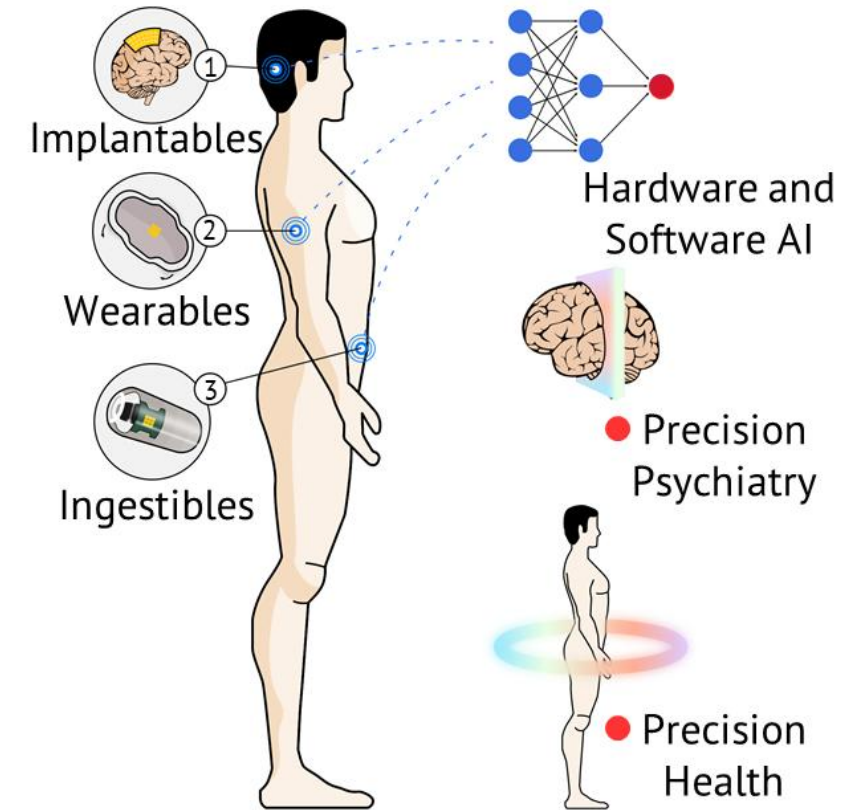




# Soft sensor systems for precision health and psychiatry



Materials,  
Manufacturing, Design,  
Devices, and Circuits



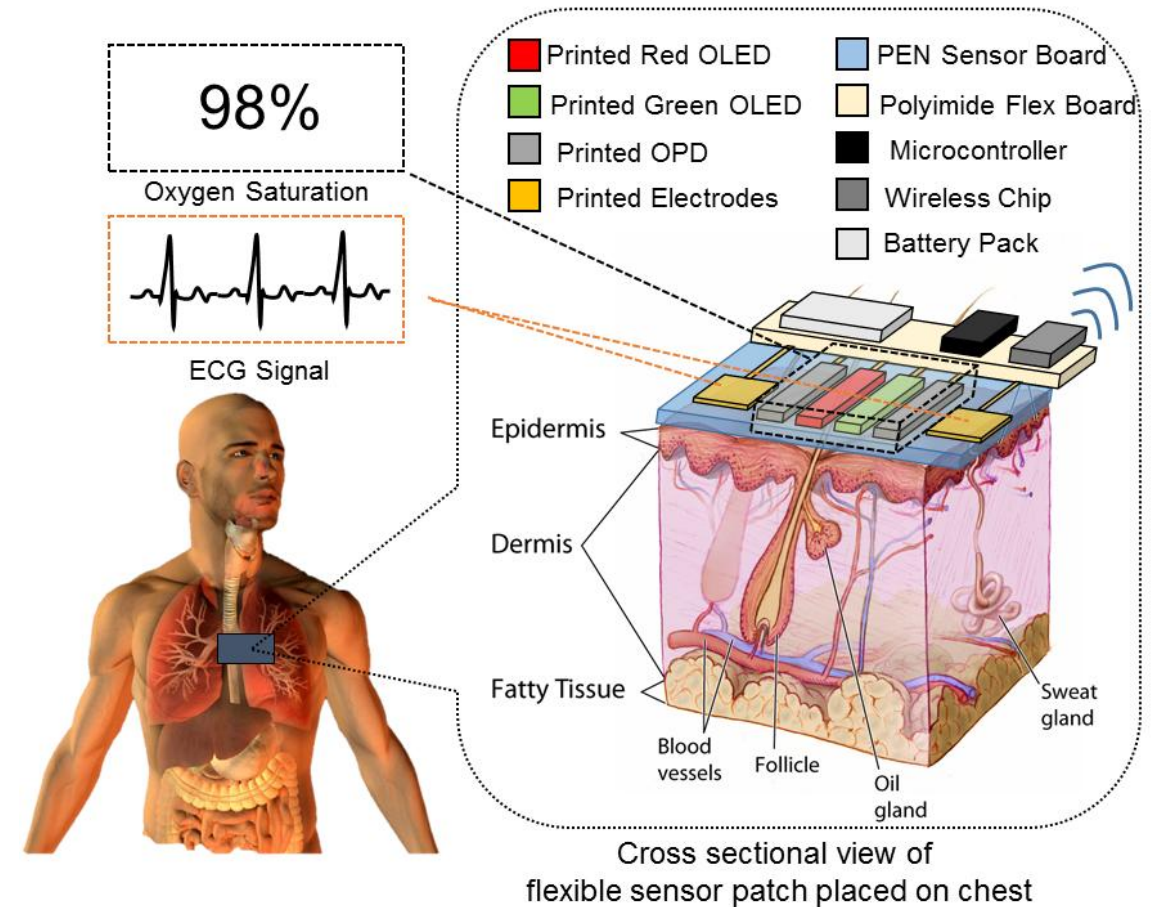
Systems and  
Applications

# Sensors for precision health and psychiatry



## Current state:

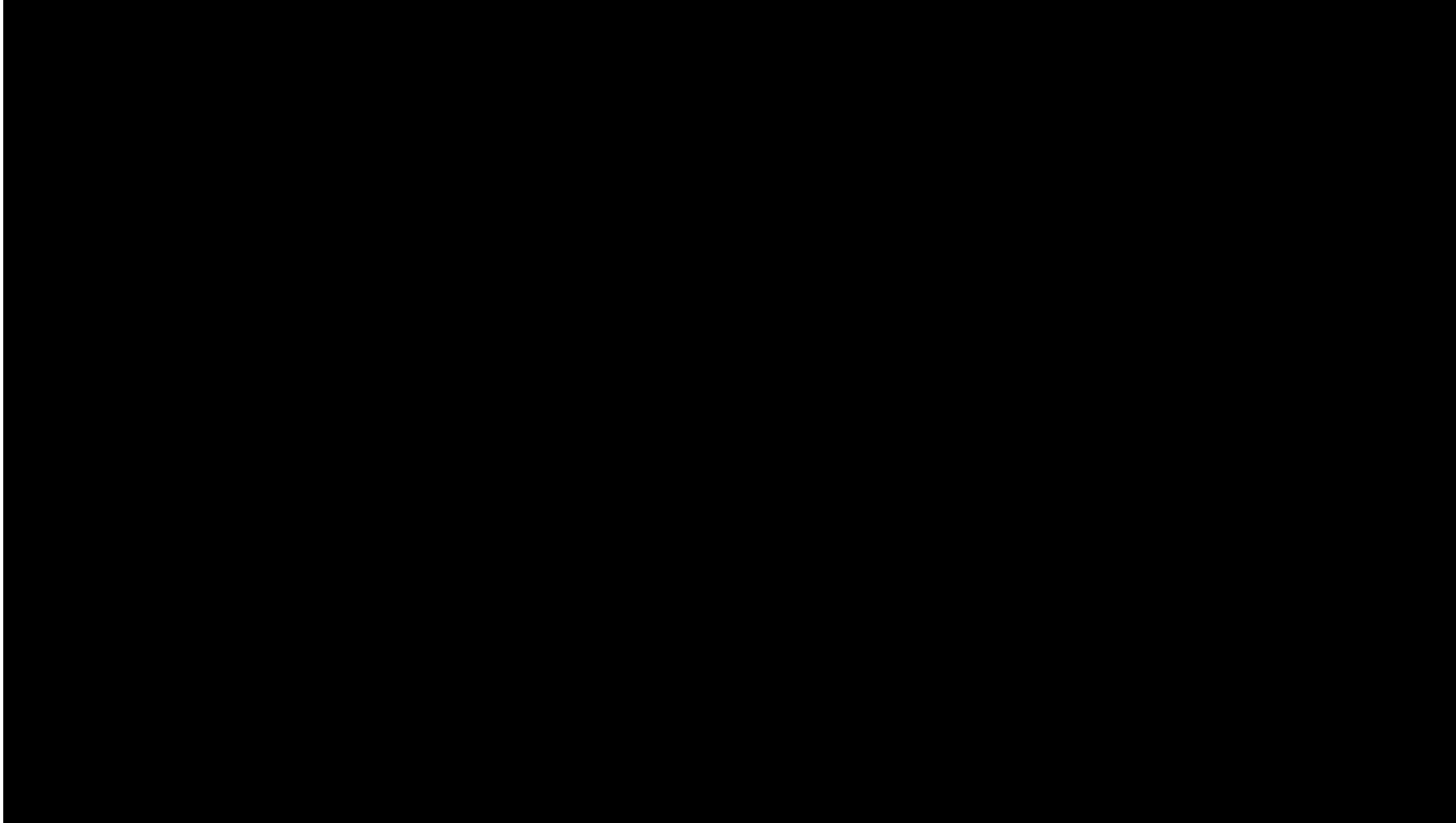
Big machines with a lot of wired rigid sensors that limits the comfort of the patient



## Vision:

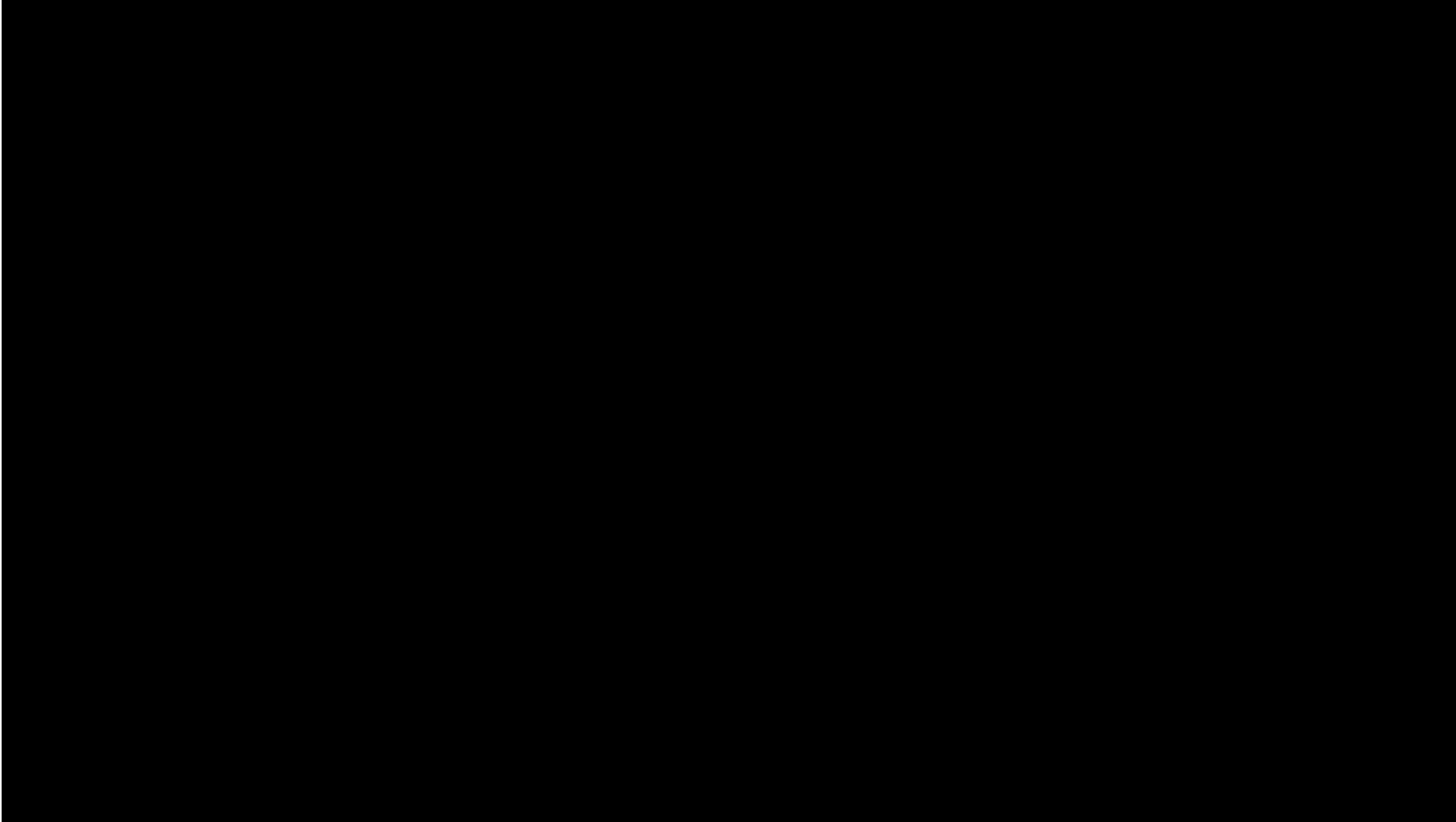
A flexible sensor patch that seamlessly measure vital signs - measurements of the body's most basic functions

# AI-powered biosensor for detecting hand gestures



- Biopotential sensing for gesture recognition

# Screen-printed MRI receive coils

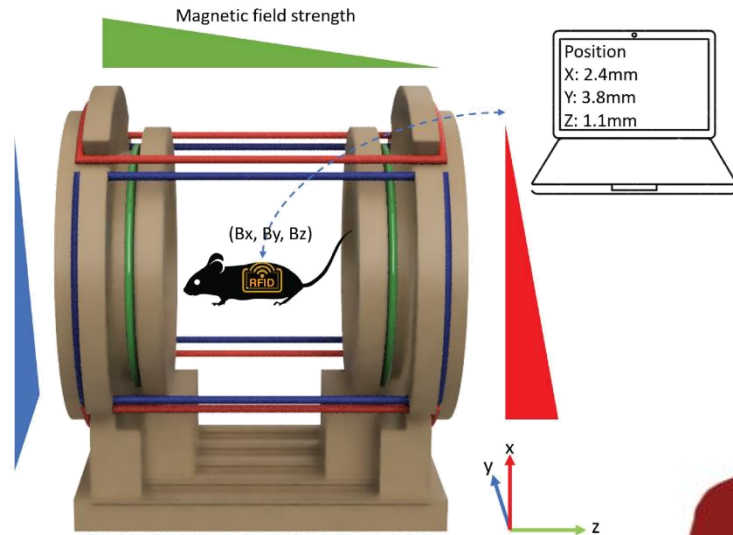


- High-throughput.
- Scalable.
- Fast.
- Significantly less process steps compared to photolithography.

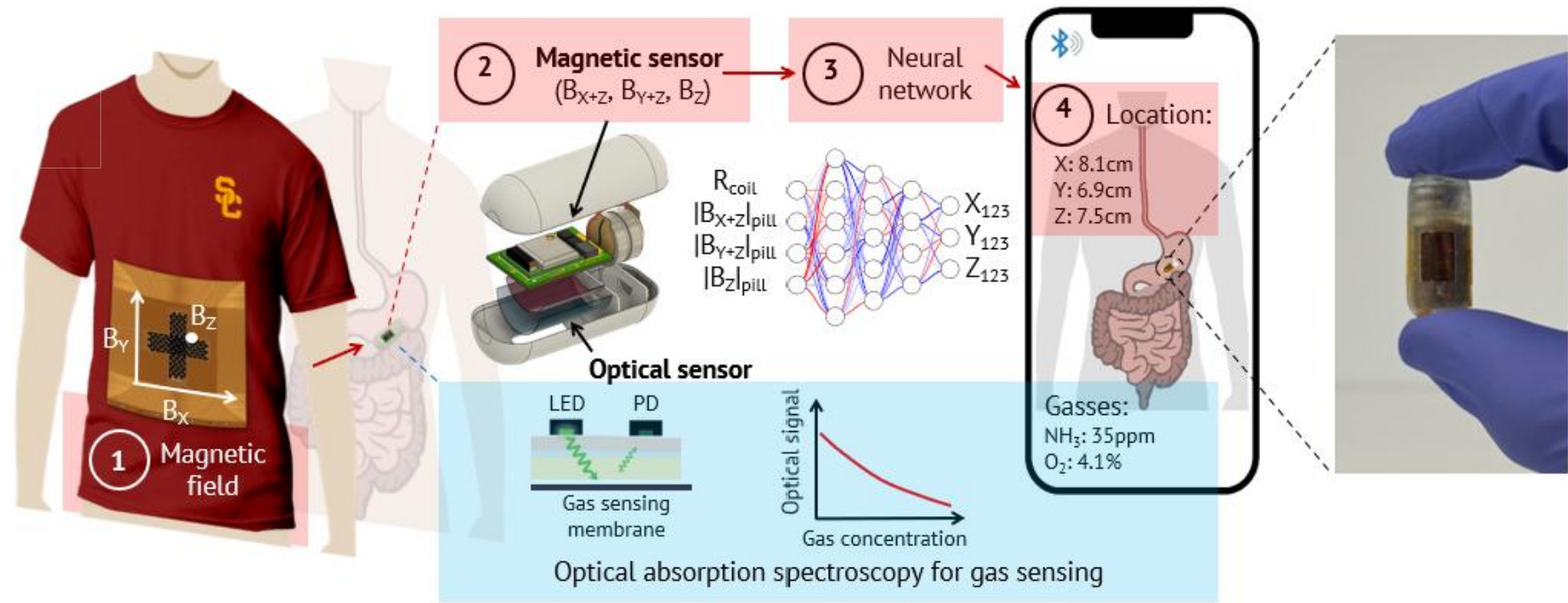


# 3D gas mapping in the gut with AI-enabled ingestible and wearable electronics

Existing approach: Helmholtz coil for producing a region of nearly uniform magnetic field and localizing ingestible inside the coil.



Our approach:  
Wearable multi-layer  
coil for 3D field  
gradient, and use a  
bend sensor for  
accounting for  
movement.



# 3D gas mapping in the gut with AI-enabled ingestible and wearable electronics

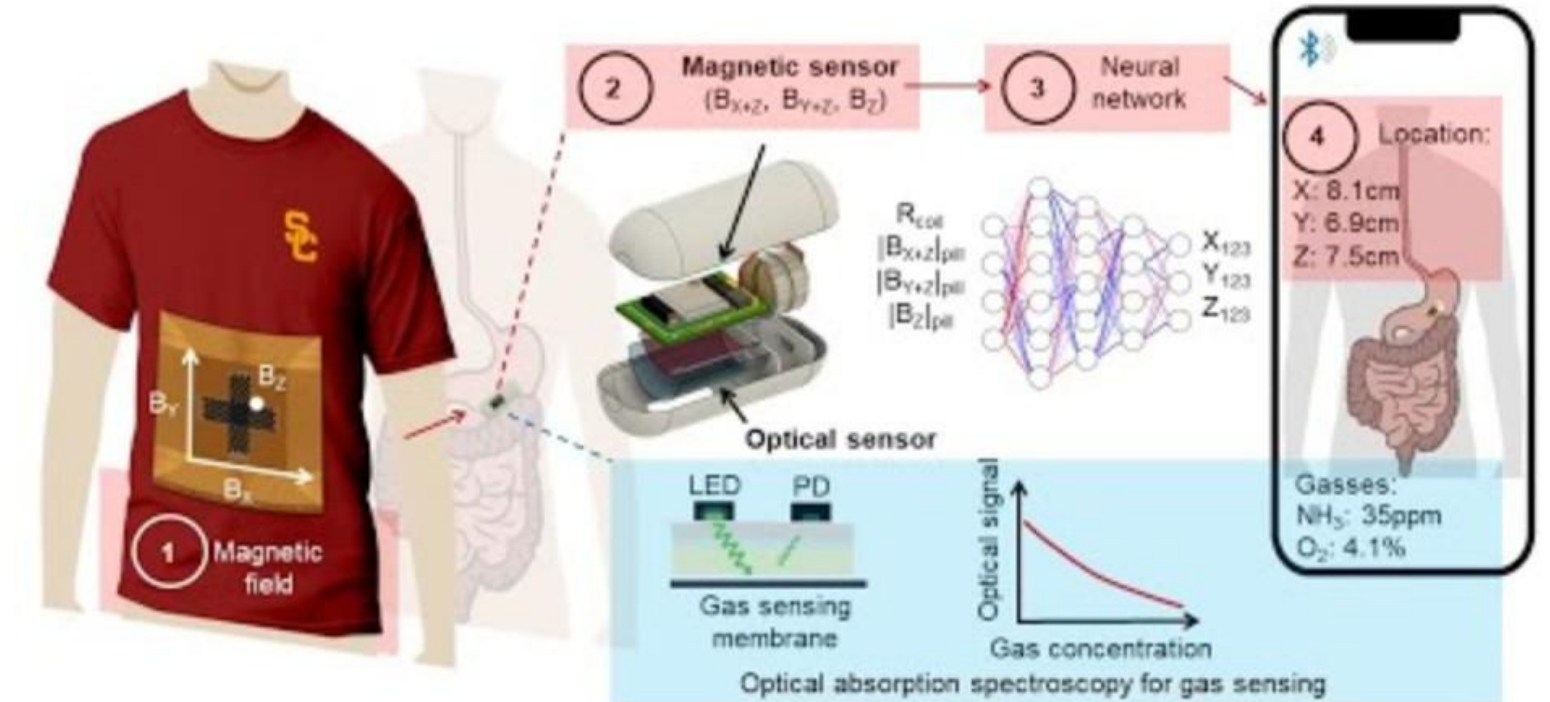
## From Wearables to Swallowables: USC Engineering Researchers Create GPS-like Smart Pills with AI

Amy Blumenthal | June 12, 2024

Utilizing wearable electronics and AI, new ingestible sensors provide real-time 3D monitoring of gastrointestinal health

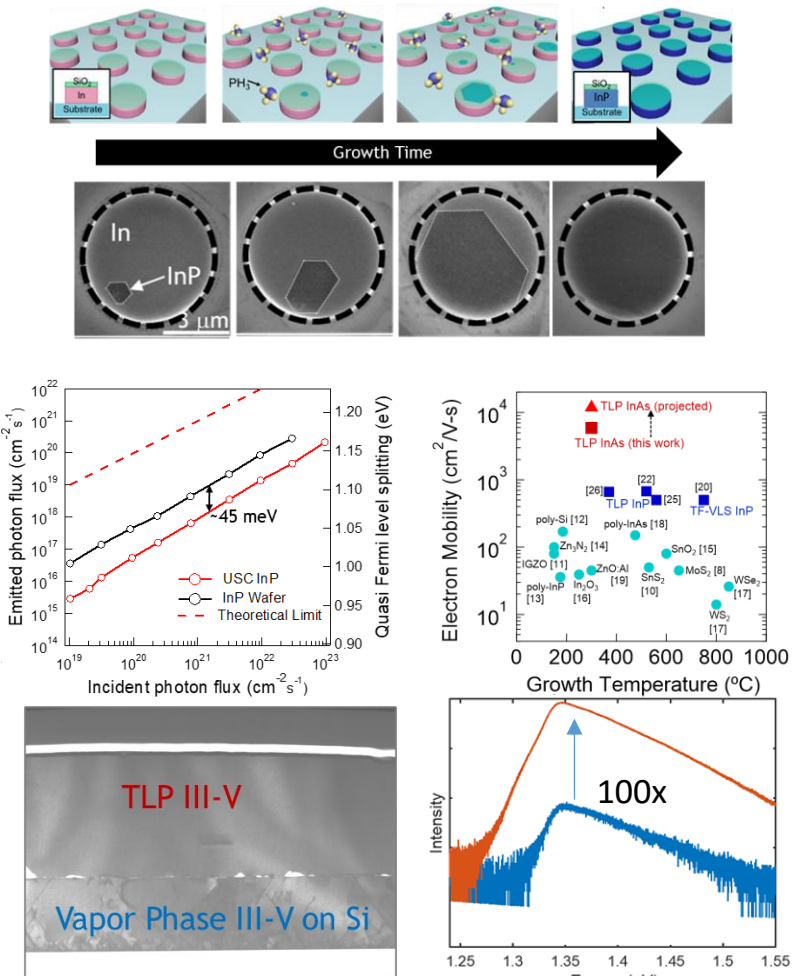


AN INGESTIBLE SMART SENSOR COULD DETECT CANCERS. IMAGE CREDIT: THE KHAN LAB

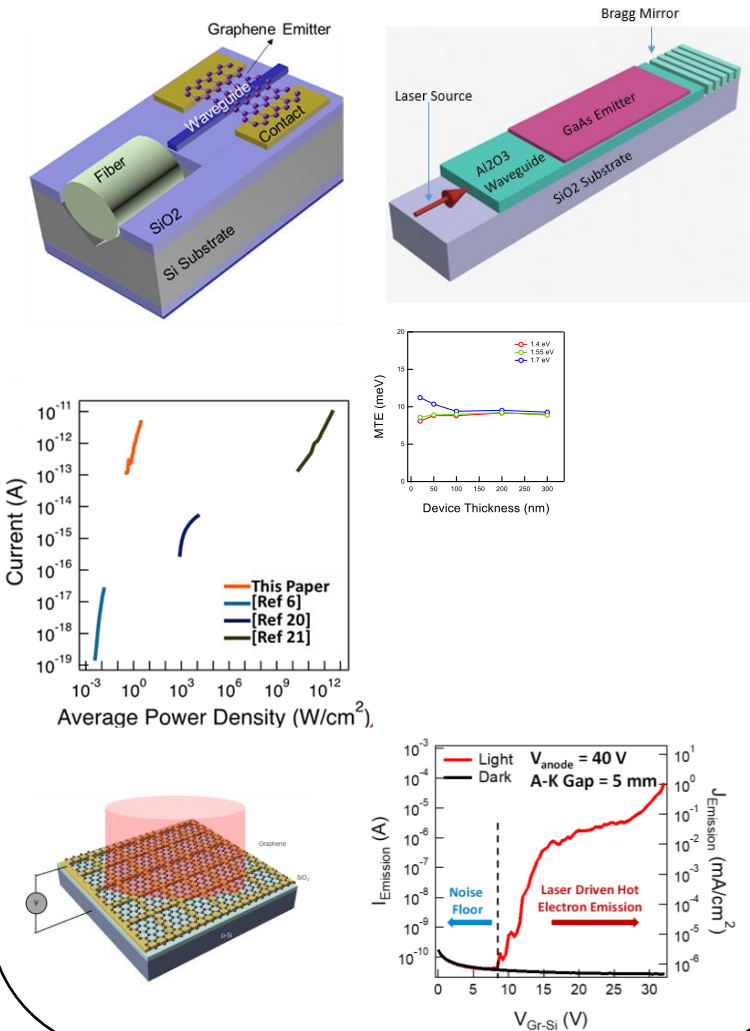


<https://youtu.be/nmVMDQnRuhw>

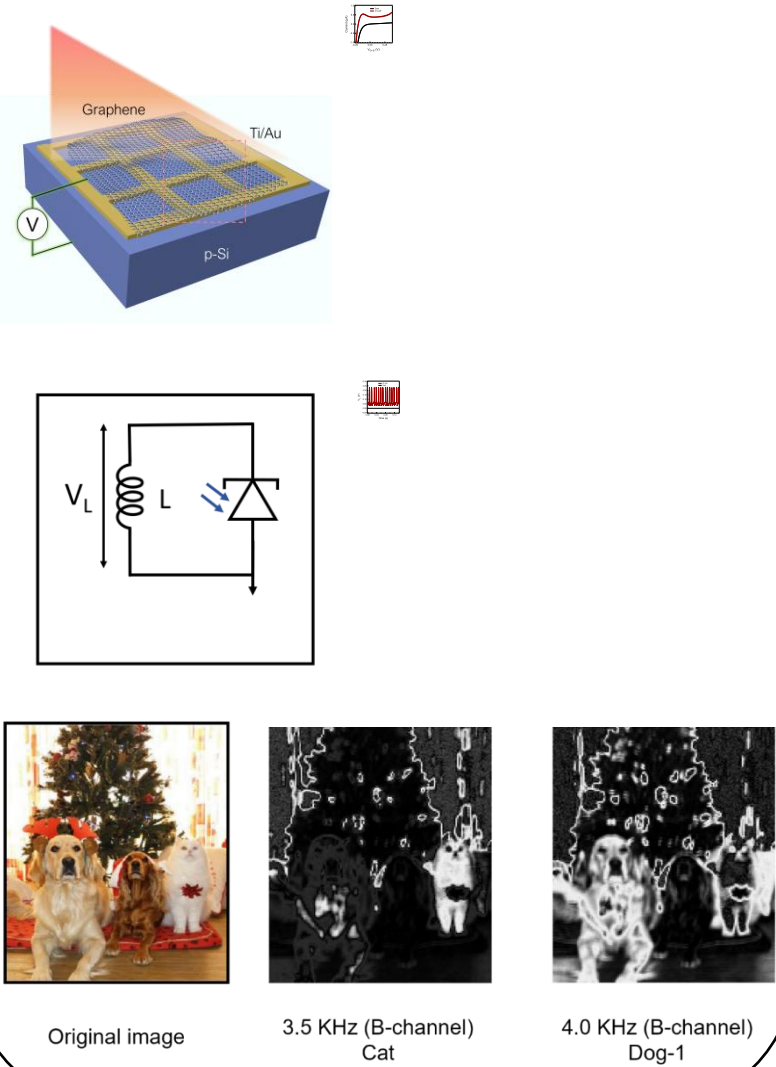
### Back-End Materials Growth and Heterogeneous Integration



### Electron Emission Devices for GHz to THz Applications



### Ultra-Low Power Computing





# John O'Brien Nanofabrication Laboratory



Etch

Deposition

Lithography

Wet Processing

Packaging

Infrastructure

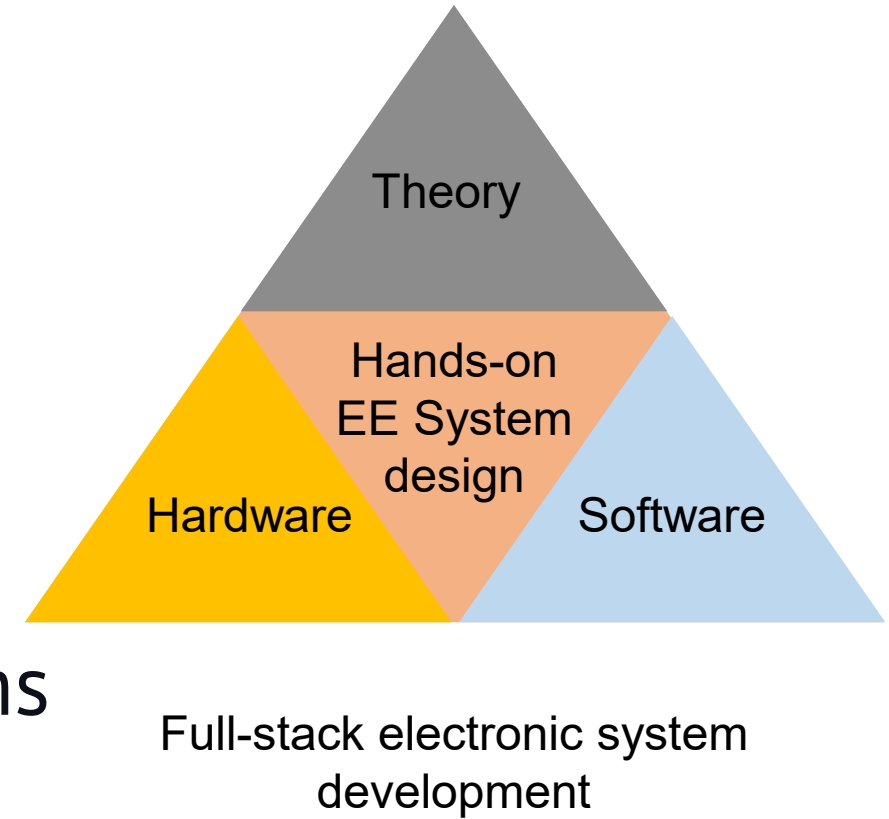
- \$50M 12,000 Sq. ft. facility
- +24,000 sq. ft. automated facility for facilities, gases, and support systems
- 35+ state-of-the-art cleanroom tools including e-beam and direct write tools, Si and compound semiconductor processing
- Sample size from pieces to 100mm (min) and 150mm (subset – but all III-V)

## Scalability:

- +200 users (2024) with expansion to +500 users planned. The lab can handle 100 active users.
- Space available within current facility for additional capacity and capability



- Understand theoretical and practical concepts in electrical engineering.
- Overview of devices, circuits, systems, and AI/ML tools.
- Design simple biosensor systems from end to end.
- Design imager systems with embedded ML.
- Design and implement electronic systems for given specifications.



- Course content and schedule: [ee105.org](http://ee105.org)
- Announcement and grading: Brightspace

## Instructor



**Yasser Khan**

[yasser.khan@usc.edu](mailto:yasser.khan@usc.edu)

I am a maker, love to build things!

## Teaching Assistants



**Shadman Shahid**

[shadmans@usc.edu](mailto:shadmans@usc.edu)

**Office Hours** Somewhere on the internet

My research is on plasmonic lasers, metasurfaces, and deep learning.



**Walter Unglaub**

[unglaub@usc.edu](mailto:unglaub@usc.edu)

I also enjoy teaching/tutoring, app development, entrepreneurship, and learning piano in my free time.

# Course structure

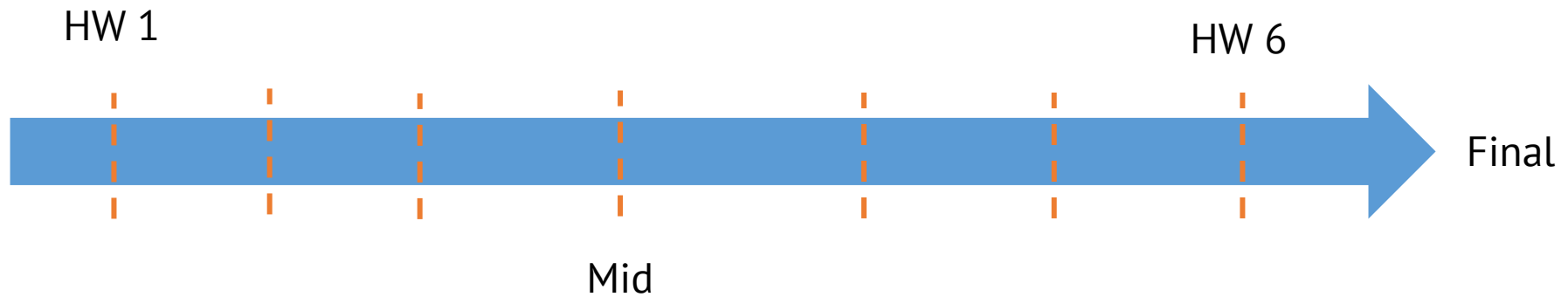
- Lectures and discussions are sufficient for homework and final exam
- Emphasis on *practical learning*
- Homework to review major concepts

Grading Breakdown

| Component     | Weight | Details                                                                                                                                                                                                                                                    |
|---------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Final Exam    | 30%    | Comprehensive; two pages of handwritten notes allowed (no printing).                                                                                                                                                                                       |
| Midterm Exam  | 30%    | Covers circuits and devices; one page of handwritten notes allowed (no printing).                                                                                                                                                                          |
| Homework      | 20%    | Six assignments:<br>1. Arduino/Python<br>2. Circuits<br>3. Devices<br>4. Sensors<br>5. Neural Networks<br>6. Optical Communications                                                                                                                        |
| Lab           | 10%    | Five reports:<br>1. Circuits<br>2. Devices<br>3. Sensors<br>4. Neural Networks<br>5. Optical Communications                                                                                                                                                |
| Final Project | 10%    | Group project (teams of 5) using demo board and Arduino.<br>• Project ideas due by <b>Oct 23</b> via email.<br>• Deliverables:<br>1. Class presentation/2-min youtube video (5 min) – 5%<br>2. Report (2 pages) – 2.5%<br>3. Source code (zip file) – 2.5% |

# Approximate schedule

- Course homepage: [ee105.org](http://ee105.org)
- 6 homework assignments
- Final in finals week
- Mid Feb 26 in class





# Course content

- 6 homework are on 6 modules
  - Arduino/python
  - Circuits
  - Devices
  - Sensors
  - Neural network
  - Communication systems
- Midterm will cover circuits and devices.
- Final will cover everything.

## Module 0 - Introduction to EE 105

|         |                                                        |                                                   |
|---------|--------------------------------------------------------|---------------------------------------------------|
| Jan 13: | <a href="#">Introduction to Electrical Engineering</a> | <a href="#">Handout the boards</a>                |
| Jan 15: | <a href="#">Introduction to Electronic Systems</a>     | <b>LAB</b> <a href="#">Show demo of the board</a> |
| Jan 20: | <b>LAB</b> <a href="#">Arduino Review</a>              |                                                   |
| Jan 22: | <b>LAB</b> <a href="#">iPython Review</a>              |                                                   |
|         | <b>HW 1 RELEASED</b>                                   | <a href="#">HW 1</a>                              |

## Module 1 - Circuits

|         |                                                  |                                                         |
|---------|--------------------------------------------------|---------------------------------------------------------|
| Jan 27: | <a href="#">Circuits 1</a>                       |                                                         |
| Jan 29: | <b>LAB</b> <a href="#">Asymmetric touchpanel</a> |                                                         |
| Feb 3:  | <a href="#">Circuits 2</a>                       |                                                         |
|         | <b>HW 1 DUE</b>                                  |                                                         |
| Feb 5:  | <a href="#">Circuits 3</a>                       | <b>LAB</b> <a href="#">Circuit analysis with python</a> |
| Feb 10: | <b>TUTORIAL</b> <a href="#">Circuit problems</a> | <b>LAB</b> <a href="#">Circuit analysis with python</a> |
|         | <b>HW 2 RELEASED</b>                             | <a href="#">HW 2</a>                                    |

## Module 2 - Device

|         |                                                |                                  |
|---------|------------------------------------------------|----------------------------------|
| Feb 12: | <a href="#">Device 1</a>                       |                                  |
| Feb 17: | <a href="#">Device 2</a>                       |                                  |
|         | <b>HW 2 DUE</b>                                |                                  |
|         | <b>HW 3 RELEASED</b>                           | <a href="#">HW 3</a>             |
| Feb 19: | <b>LAB</b> <a href="#">Diode analysis</a>      |                                  |
|         | <b>LAB</b> <a href="#">Transistor analysis</a> |                                  |
| Feb 24: | <a href="#">Device 3</a>                       |                                  |
|         | <b>PRACTICE MIDTERM</b>                        | <a href="#">Practice MIDTERM</a> |
|         | <b>HW 3 DUE</b>                                |                                  |
| Feb 26: | <b>MIDTERM</b>                                 |                                  |

# Course content

### Module 3 - Sensors

|         |                                                                    |                                                   |
|---------|--------------------------------------------------------------------|---------------------------------------------------|
| Mar 3:  | <a href="#">Sensors 1</a>                                          | <b>LAB</b> <a href="#">Temperature sensor</a>     |
| Mar 5:  | <a href="#">Sensors 2</a>                                          | <b>FINAL PROJECT TEAMING</b> Groups of 5 students |
| Mar 10: | <b>LAB</b> <a href="#">Oximeter data collection and processing</a> |                                                   |
|         | <b>LAB</b> <a href="#">Lab files</a>                               |                                                   |
| Mar 12: | <a href="#">Sensors 4</a>                                          |                                                   |
|         | <b>FINAL PROJECT PLAN DUE</b>                                      |                                                   |
|         | <b>HW 4 RELEASED</b> <a href="#">HW 4</a>                          |                                                   |

### Module 4 - Machine Learning

|         |                                                       |                                                                               |
|---------|-------------------------------------------------------|-------------------------------------------------------------------------------|
| Mar 24: | <a href="#">Nueral network 1</a>                      |                                                                               |
| Mar 26: | <a href="#">Nueral network 2</a>                      | <b>LAB</b> <a href="#">Iimage/text classification</a>                         |
| Mar 31: | <a href="#">Nueral network 3</a>                      | <b>LAB</b> <a href="#">Iimage/text classification</a>                         |
| Apr 2:  | <a href="#">Nueral network 4</a>                      | <b>LAB</b> <a href="#">Iimage/text classification with arduino and TinyML</a> |
|         | <b>HW 4 DUE</b>                                       |                                                                               |
|         | <b>HW 5 RELEASED</b> <a href="#">HW 5</a>             |                                                                               |
| Apr 7:  | <b>LAB</b> <a href="#">Iimage/text classification</a> |                                                                               |

### Module 5 - Communications

|         |                                                          |                                           |
|---------|----------------------------------------------------------|-------------------------------------------|
| Apr 9:  | <a href="#">Communications 1</a>                         |                                           |
| Apr 14: | <b>LAB</b> <a href="#">LiFi</a>                          |                                           |
| Apr 16: | <a href="#">Communications 2</a>                         | <b>DEMO</b> <a href="#">Sensor fusion</a> |
| Apr 21: | <b>NO CLASS - THANKSGIVING</b>                           |                                           |
|         | <b>HW 6 RELEASED</b> <a href="#">HW 6</a>                |                                           |
|         | <b>PRACTICE FINAL</b> <a href="#">Practice FINAL</a>     |                                           |
|         | <b>FINAL TOPIC LIST</b> <a href="#">FINAL topic list</a> |                                           |
|         | <b>HW 5 DUE</b>                                          |                                           |
| Apr 23: | <b>NO CLASS - PROJECT WORK</b>                           |                                           |

### Module 6 - Final Project Presentation / Demo

|         |                 |  |
|---------|-----------------|--|
| Apr 28: | <b>TEAM 1</b>   |  |
|         | <b>TEAM 2</b>   |  |
|         | <b>TEAM 3</b>   |  |
|         | <b>TEAM 4</b>   |  |
| Apr 30: | <b>TEAM 5</b>   |  |
|         | <b>TEAM 6</b>   |  |
|         | <b>TEAM 7</b>   |  |
|         | <b>TEAM 8</b>   |  |
|         | <b>TEAM 9</b>   |  |
|         | <b>HW 6 DUE</b> |  |

# Custom demonstration board for EE 105

Circuits

Voltage dividers

Arduino Nano 33  
BLE

Electronic devices  
Diodes  
Transistors

Module 2: Devices

Electronic systems  
Oximeter

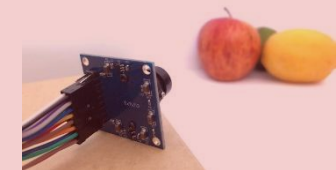
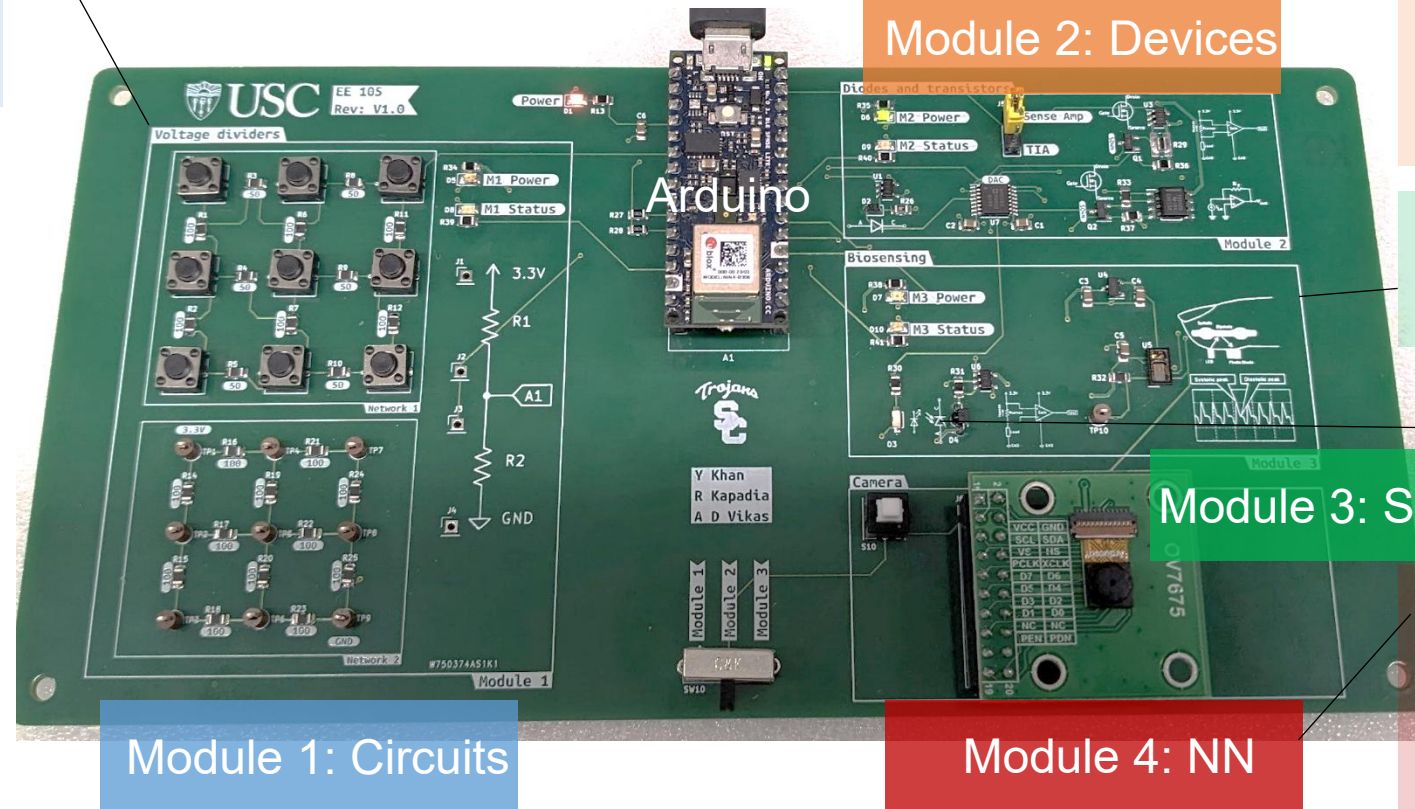
Module 5: Comm

Module 3: Sensors

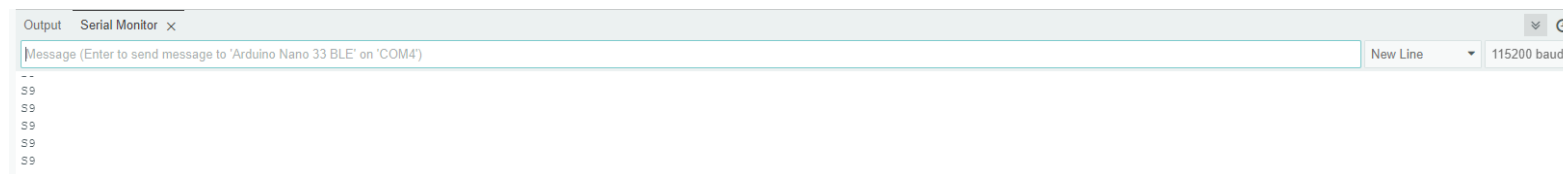
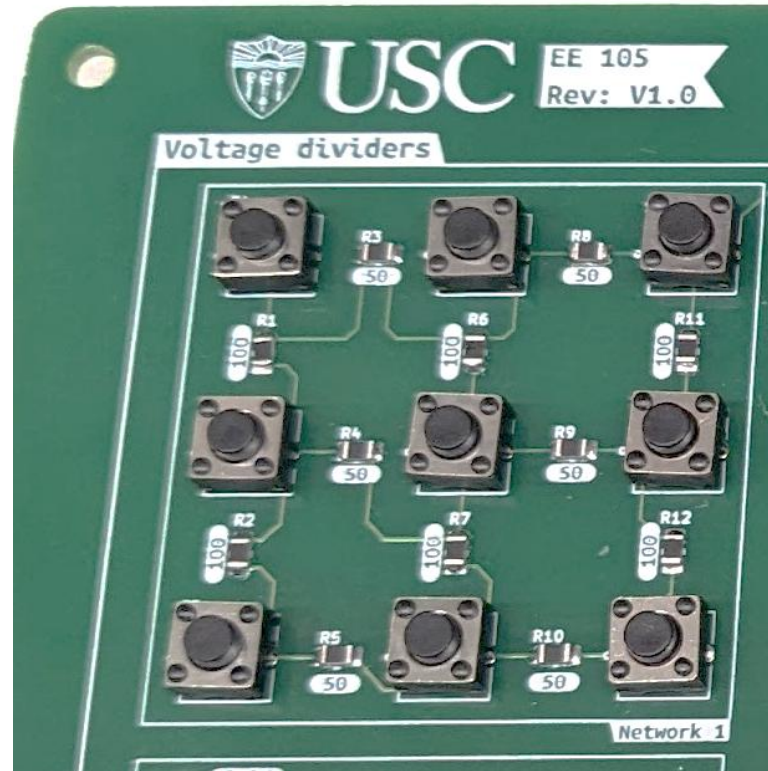
Machine vision

Module 4: NN

Module 1: Circuits



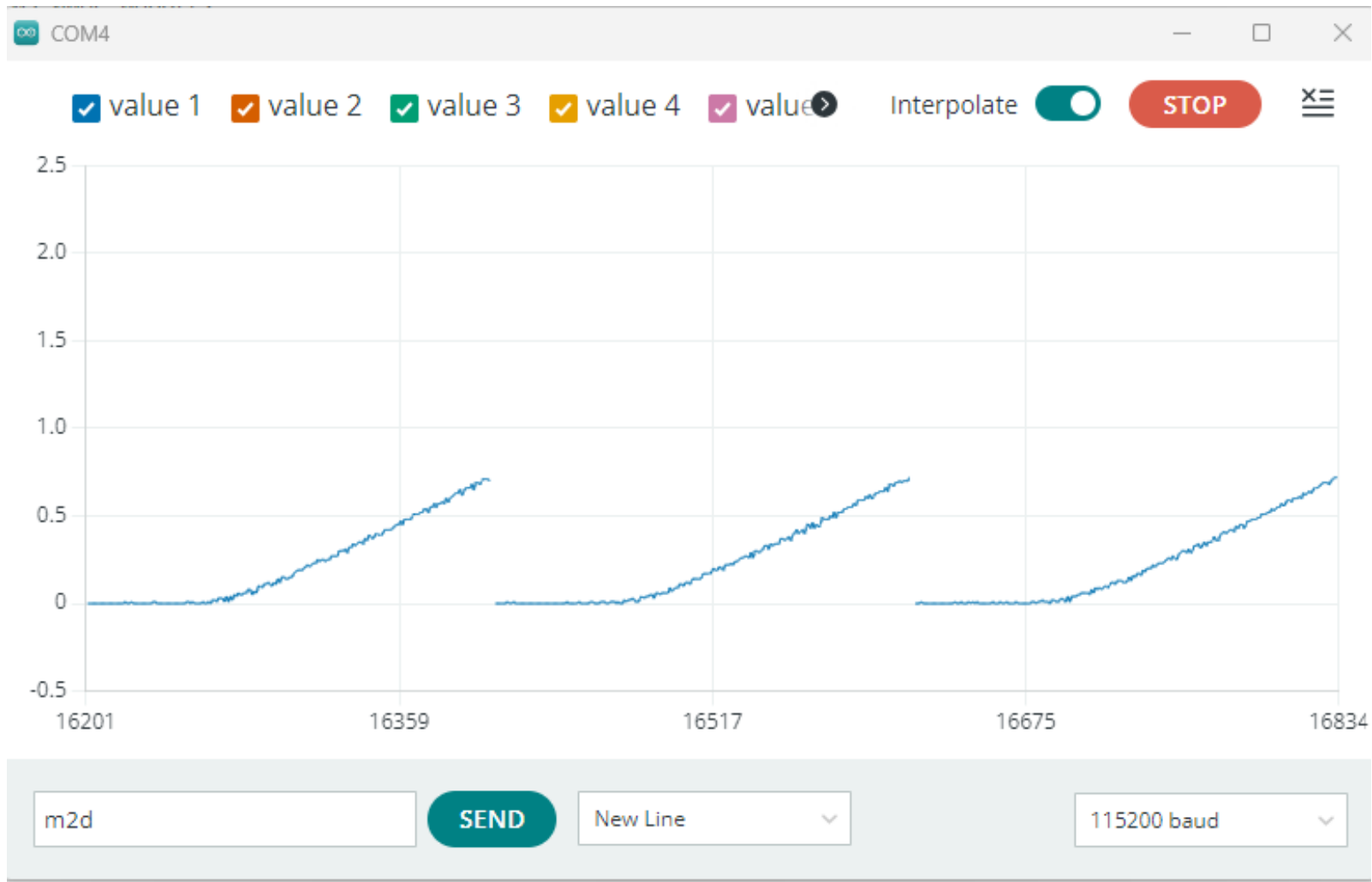
# Module 1 on circuits – demo resistive touchscreen



Finds the pressed button location automatically

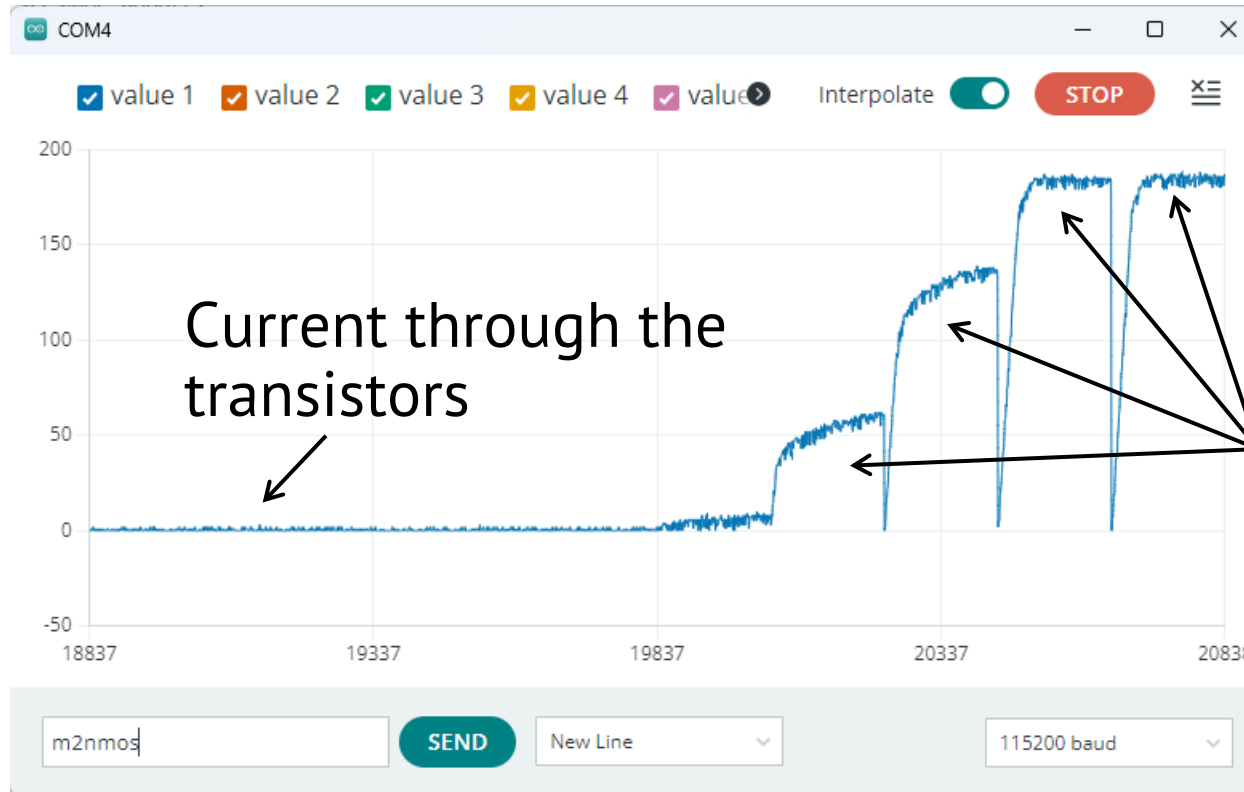


# Module 2: diodes and transistors



Characterize  
semiconductor  
diode – how do  
they work

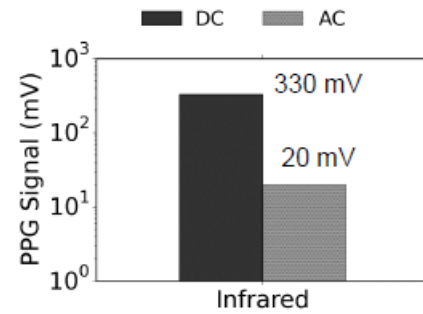
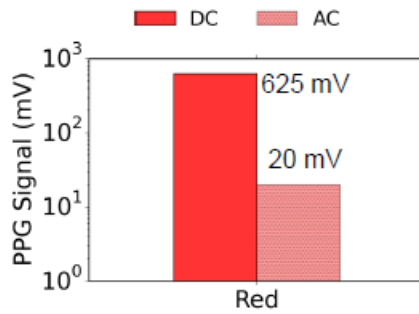
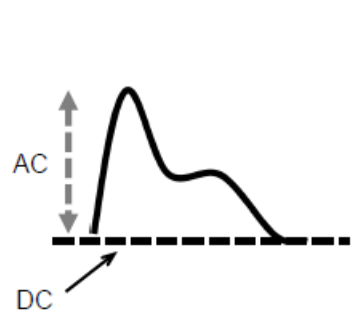
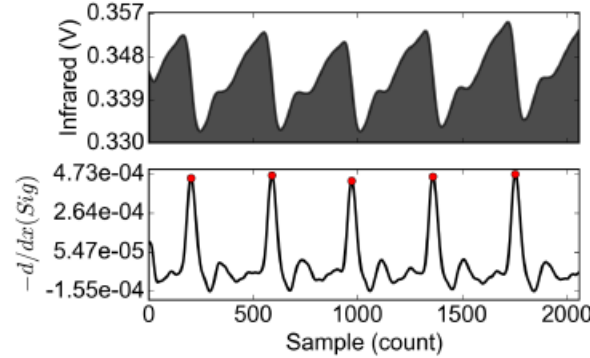
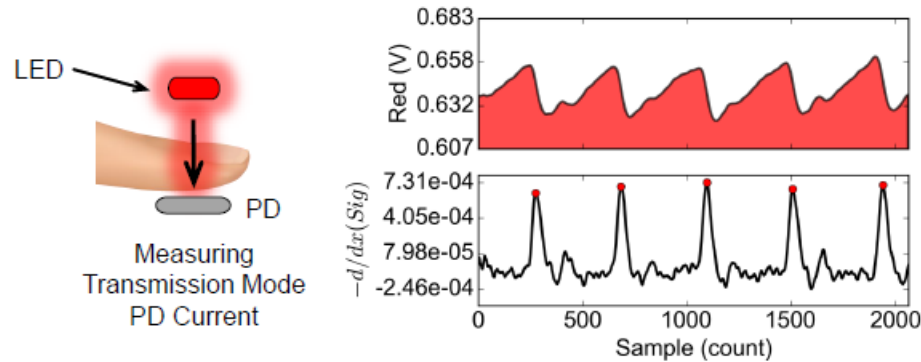
# Module 2: diodes and transistors



Characterize  
semiconductor  
transistors – how  
do they work

Increasing gate  
voltage / opening  
faucet in the water  
analogy

# Module 3: sensors

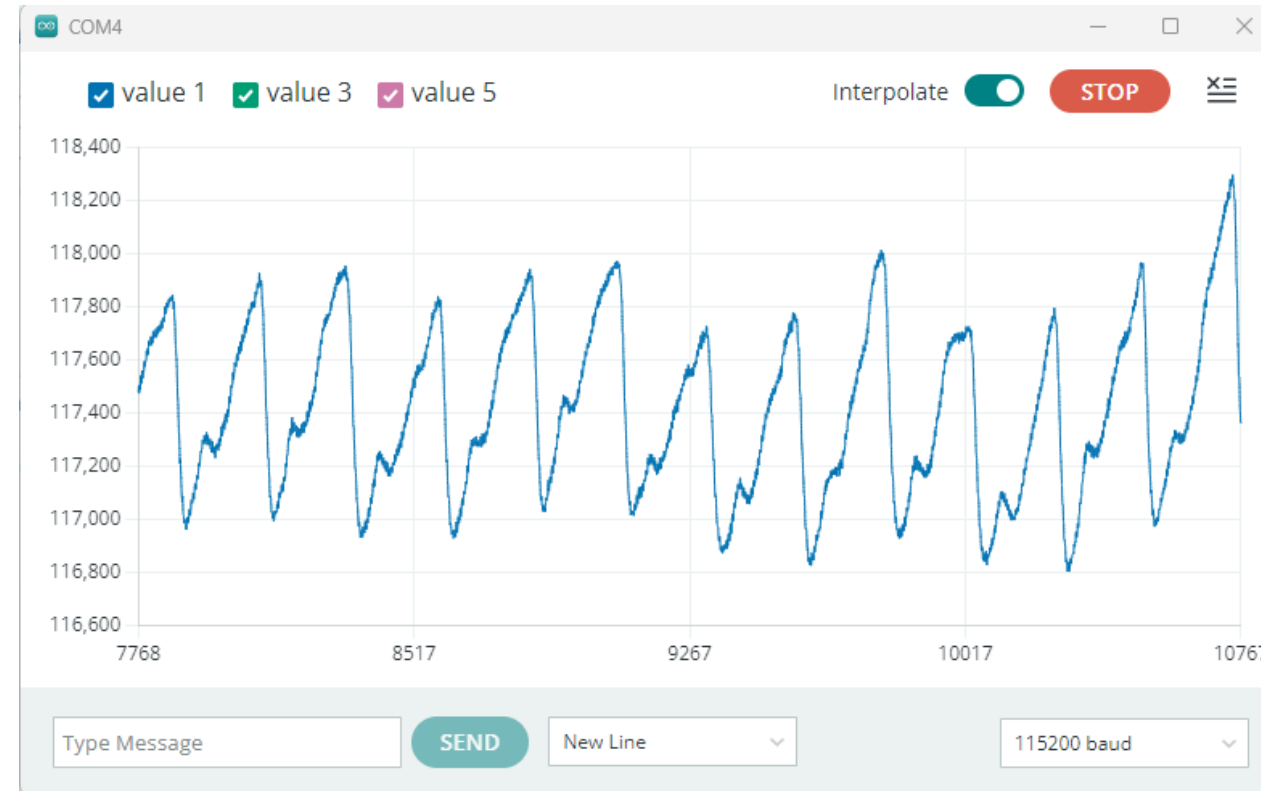


$$R = \frac{AC_{rd}/DC_{rd}}{AC_{ir}/DC_{ir}}$$

$$R = \frac{20\text{mV}/625\text{mV}}{20\text{mV}/330\text{mV}} = .528$$

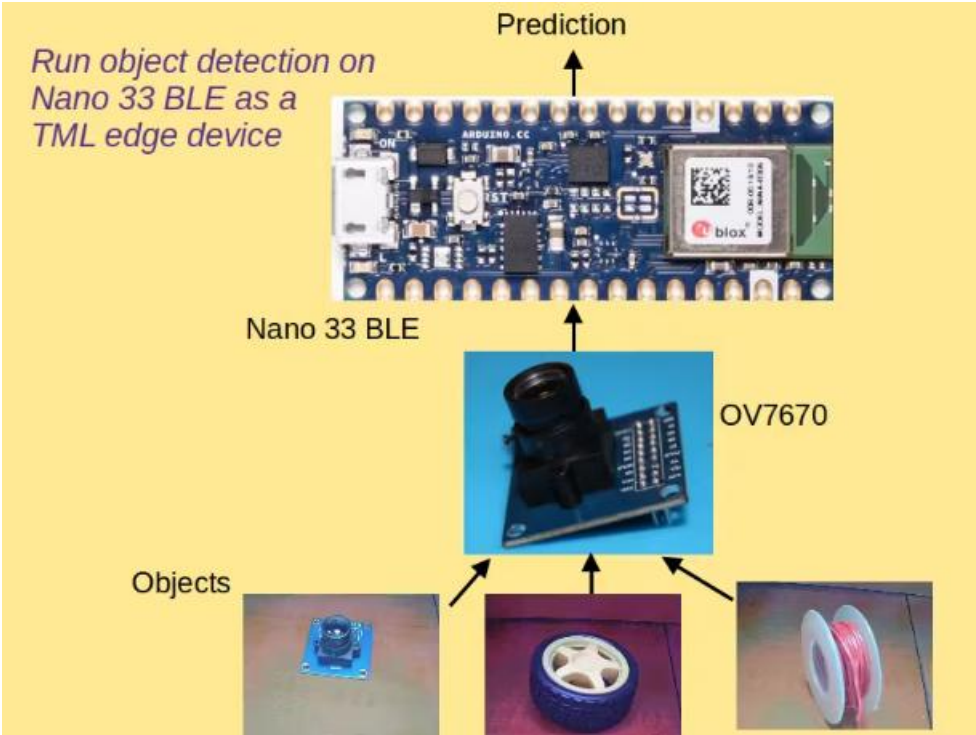
$$SpO_2 = 110 - 25 * R = 97\%$$

# Module 3: sensors - oximeter

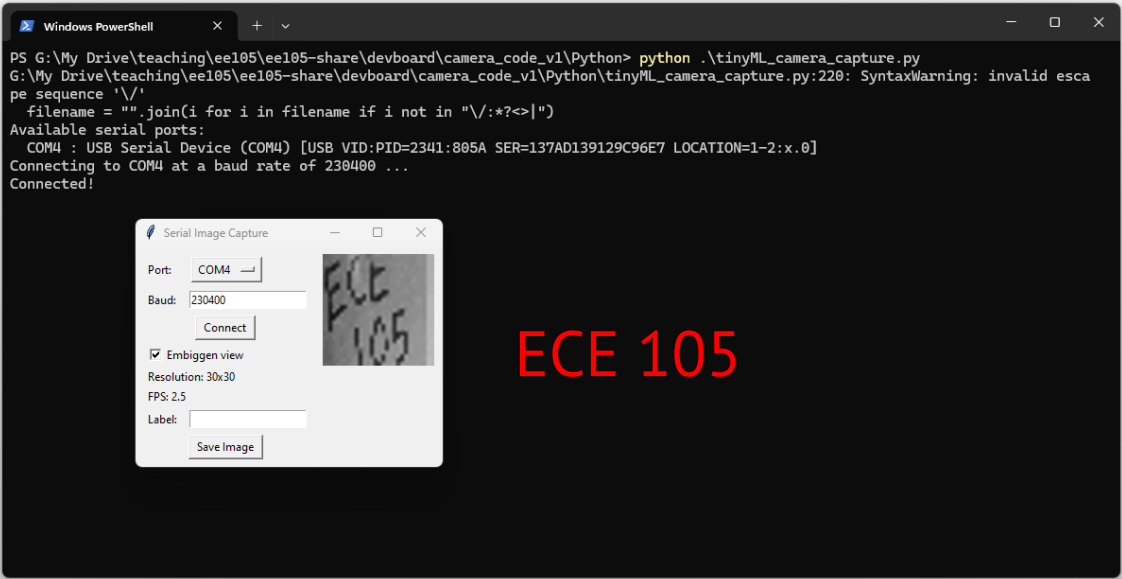
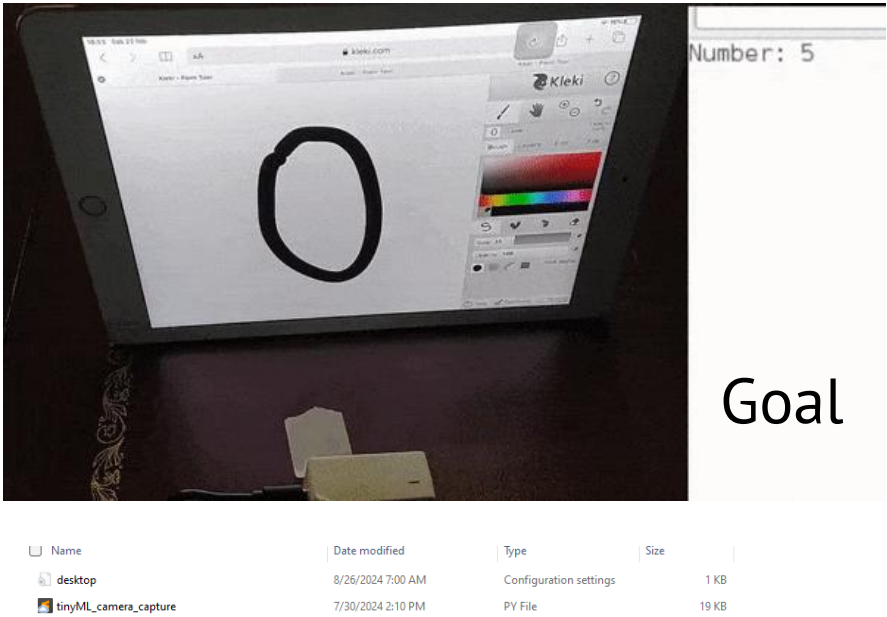




# Module 4: Machine vision

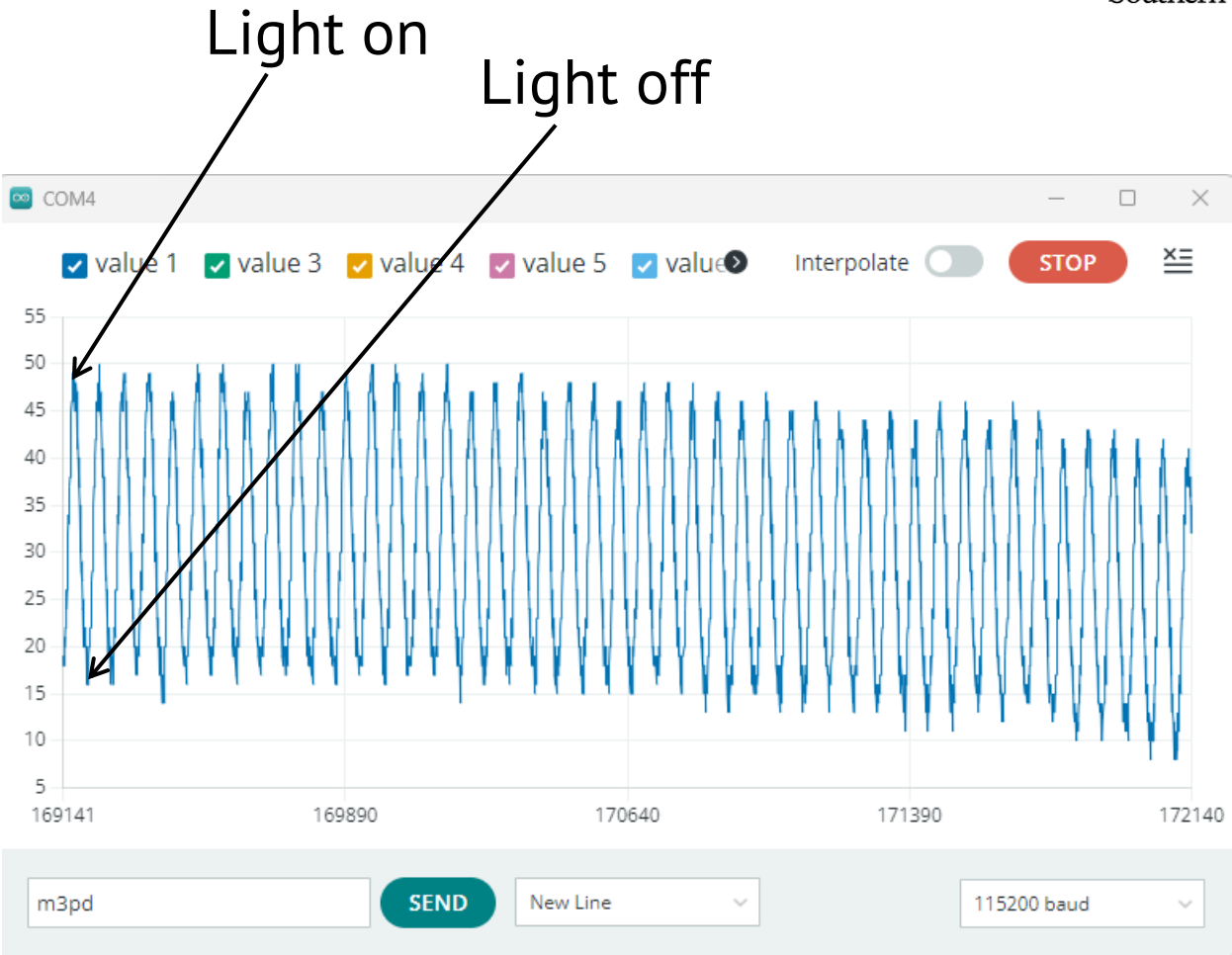
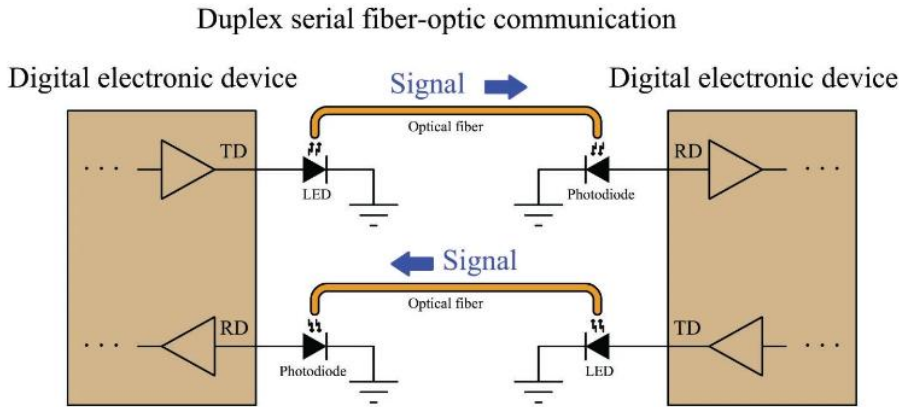


Arduino + camera + ML =  
identify objects / letters /  
faces / anything



ECE 105

# Module 5: optical communication



# Next lecture: system design

